

From Hypothesis to Evidence: Evidence-Carrying AI-Assisted Cryptanalysis for Post-Quantum Forensic Assurance

A verification-graded framework, formal evidence model, and evaluation protocol

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Abstract

During 2026, artificial intelligence (AI) moved from assisting cryptanalysts to proposing and testing cryptanalytic attacks itself. An AI-assisted key-recovery attack roughly halved the security of the HAWK signature scheme, which its authors then withdrew from NIST’s additional-signature process [1], [2], [3]. An LLM benchmark recorded novel attacks on lesser-studied schemes [4]. An autonomous workflow returned exact witnesses for flaws in eight published constructions [5]. None of these results affects the standardized parameter sets of ML-KEM, ML-DSA or SLH-DSA, and the people reporting them stress that verifying the results, not generating them, is now the bottleneck [6].

This paper asks how AI-assisted cryptanalysis can stop being an unreliable source of hypotheses and become reproducible, independently verifiable, forensic-grade evidence for post-quantum risk assessment and migration. We make four contributions.

1. **A claim taxonomy and verification ladder.** Every AI-generated cryptanalytic claim is classified by its logical form (existential witness, statistical advantage, complexity bound, proof defect, implementation fault, protocol flaw or negative result) and by its scope (design, reduced-parameter, implementation, protocol or assistance). It is then graded on four separate assurance levels: hypothesis, executable support, independent reproduction and formal verification. These levels are never merged into one confidence score.
2. **An evidence object $C = (H, W, E, V, P)$.** It binds a scoped hypothesis to its witness, its experiment record, its verification records and its provenance. Its integrity is preserved in an append-only, signed, Merkle-anchored log with algorithm-agile signatures.
3. **A cryptographic digital twin with an HNDL risk model.** The risk model separates *retrospective* exposure (data already harvested, which migration cannot fix) from *prospective* exposure (ongoing flows, which migration can fix). Its parameters are probabilities and elicited sensitivities, not free weights.
4. **A pre-registered evaluation protocol.** It specifies a benchmark with planted ground truth and hallucination traps, eleven metrics including a new Scope-Inflation Rate, six baselines and a seven-way ablation.

We report no experimental results and make no claim that any standardized post-quantum scheme is weakened. We identify which parts of the framework are new, which integrate existing techniques, and which must be demonstrated experimentally before they can be claimed.

Keywords: AI-assisted cryptanalysis; post-quantum cryptography; ML-KEM; ML-DSA; LLM agents; side-channel analysis; digital forensics; harvest-now-decrypt-later; formal verification; evidence provenance; cryptographic agility; reproducibility.

1. Introduction

1.1 Motivation

The migration to post-quantum cryptography (PQC) is now an operational programme. NIST published FIPS 203 (ML-KEM), FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) in August 2024 [7], [8], [9]. It selected the code-based KEM HQC for standardization in March 2025 [10], and it plans to deprecate and then remove quantum-vulnerable public-key algorithms from its standards by 2035 [11]. The concern behind this schedule is “harvest now, decrypt later” (HNDL). An adversary records ciphertexts today and decrypts them once a cryptographically relevant quantum computer (CRQC) exists, so data whose confidentiality must outlast that date is already exposed [12], [13].

A second, separate change happened in the same period: AI became an active participant in cryptanalysis. Three 2026 results illustrate this.

- **HAWK.** Anthropic reported that one researcher working semi-autonomously with its Claude Mythos model found a non-trivial automorphism in the lattice underlying HAWK. This enabled a faster key-recovery attack that roughly halves the effective security. For the smallest parameter set, HAWK-256, the expected attack cost fell from about 2^{64} to about 2^{38} [1]. The HAWK team withdrew the scheme from NIST’s additional-signature process on 29 July 2026 [2], [3].
- **CryptanalysisBench.** On a benchmark of 191 schemes, five frontier models broke 65–86% of the 49 known-broken schemes. They also produced novel results, including a key-recovery attack on the SpoC AEAD and an error in the CCA-security proof of KINDI [4].
- **AI Grinding.** An autonomous “AI grinding” workflow returned exact witnesses and controls for failures in eight published constructions [5].

These results do **not** show that standardized PQC is broken. The HAWK attack is specific to HAWK and does not affect other NIST candidates or lattice cryptography generally [1]. The 7-round AES result announced alongside it is a constant-factor improvement on a reduced-round variant under a chosen-plaintext model [1], [6]. The results do change the economics of cryptanalysis: generating hypotheses is becoming cheap. Commenting on these results, Green argues that verification is now the bottleneck [6]. In the AES case, human validation took far longer than the model’s discovery [1].

1.2 The problem: unverified claims enter security and forensic decisions

AI-generated cryptanalytic claims are now reaching migration planners, incident responders and forensic examiners. Three properties make them hazardous in those settings:

1. **They can be wrong in ways that look right.** Fluent output can describe attacks that are invalid, that apply only to a weakened instance, or that confuse an implementation bug with a mathematical break. LLMs are much better at interpreting and completing formal specifications than at writing or correcting them from scratch [14].
2. **Confirmation is often not independent.** If two analysts “confirm” a claim with the same model, prompt template or codebase, they may share the same error. Commentary on the 2026 results calls this the *independence problem* [15].
3. **Forensic and legal use needs more than correctness.** It needs integrity, chain of custody, reproducibility, tool validation and explainable methods [16], [17]. In the United States, a proposed evidence rule aimed specifically at machine-generated evidence was returned for revision in June 2026, which shows that legal standards here are still unsettled [18].

1.3 Research question

***RQ*:** How can AI-assisted cryptanalysis be turned from a potentially unreliable hypothesis-generation mechanism into a reproducible, independently verifiable, forensic-grade capability that supports post-quantum risk assessment and cryptographic migration?*

We assess the following proposition rather than assume it:

AI changes the cryptanalytic threat landscape by accelerating hypothesis generation, attack composition, implementation analysis and vulnerability discovery. Post-quantum mitigation therefore requires not only stronger algorithms but also mechanisms for evaluating, verifying and forensically preserving AI-assisted cryptanalytic findings.

Section 15.1 examines the evidence for and against this proposition. In brief, the first sentence is well supported for implementation analysis and for lesser-studied designs. There is no evidence that it holds for the design-level security of the standardized schemes. The second sentence follows whenever AI-generated claims influence decisions, whatever their real hit rate.

1.4 Contributions

Based on the literature reviewed through 30 September 2026 (Sections 3–5 and Appendix B), we make the following contributions. Each one is labelled by what kind of contribution it is.

- **C1 — Claim taxonomy and verification ladder** (*conceptual and formal; new as an integrated, claim-type-specific scheme*). Each claim is classified by its *logical form* and *scope*. The form determines the appropriate verifier (Table 9). The scope labels (design, reduced, implementation, protocol, assistance) cannot be merged. The four assurance levels are kept as separate flags.
- **C2 — The evidence object $C = (H, W, E, V, P)$ and its provenance log** (*formal; integrates existing provenance techniques [19], [20], [21], [22], [23]*). We define validity conditions, an independence relation for reproduction, and an append-only, algorithm-agile log.
- **C3 — A cryptographic digital twin and a two-component HNDL risk model** (*formal; extends Mosca’s inequality [12] and temporal HNDL models [13]*). Retrospective and prospective exposure are separated, and every parameter can be calibrated or validated.
- **C4 — An evaluation protocol and benchmark design** (*methodological; requires experimental validation*). It includes planted ground truth, hallucination traps, an independence-controlled reproduction track, eleven metrics including the new Scope-Inflation Rate, six baselines and an ablation.

We deliberately make **no** empirical contribution claim in this version. Section 14 states what must be shown before any of C1–C4 can be called effective.

1.5 Labelling conventions

The paper marks the epistemic status of each statement: **[Fact]** for an established fact with a primary source; **[Published]** for a result reported by others, which we have not reproduced; **[Hypothesis]** for our hypothesis; **[Proposed]** for our architecture or definition; **[Expected]** for an expected outcome, not a result; **[Limitation]**; and **[Open]** for an open question. Where it is unambiguous, the label is implicit in the section heading.

2. Background

2.1 Cryptanalysis and the five attack classes

A cryptanalytic result is a statement about a *target* under an *attack model* with specific *resources*. We use five mutually exclusive classes. The whole paper depends on keeping them apart.

Table 1. Attack-class taxonomy used throughout this paper.

Class	What is attacked	Example (2019–2026)	What a success does <i>not</i> imply
A. Design-level	The mathematical construction at its specified parameters	SIDH key recovery [24]; Rainbow key recovery [25]; HAWK security reduction [1]	Nothing about other schemes built on related assumptions (HAWK $\neq$ Falcon $\neq$ ML-DSA)
B. Reduced - parameter	Weakened instances (fewer rounds, smaller dimension, sparse secrets)	Neural distinguishers on round-reduced Speck [26]; SALSA-family LWE with sparse secrets [27], [28]; 7-round AES [1]	That the attack scales to full parameters
C. Implementation	A specific implementation on a specific platform (timing, power, EM, faults, bugs)	DL side-channel attacks on masked Kyber [29]; KyberSlash timing [30]; incomplete FO comparison in wolfSSL ML-KEM [31]	That the algorithm or standard is broken
D. Protocol	Negotiation, downgrade, hybrid composition, key management, misuse	Hybrid-combiner and cascade evaluations [32]	That the underlying primitive is weak

Class	What is attacked	Example (2019–2026)	What a success does <i>not</i> imply
E. Assistance	AI helps a human find, optimize or verify an attack	HAWK (human + model) [1]; CryptanalysisBench agent runs [4]	That the AI broke anything by itself

[Fact] The two famous design-level breaks of NIST round-3 and round-4 candidates (Rainbow and SIDH) were found by humans without AI [24], [25]. Classical automated tools such as MILP- and SAT-based characteristic search predate neural methods by a decade [33]. “Automated cryptanalysis” is therefore not new; what is new in 2026 is *open-ended hypothesis generation* by language-model agents.

2.2 Artificial intelligence in cryptanalysis

AI in cryptanalysis falls into three families:

1. **Learned distinguishers and key-recovery models.** These include neural differential distinguishers, which began with Gohr’s 11-round Speck32/64 key recovery [26] and were systematized by Benamira et al. [34]. A 2026 survey reviews 238 publications, 71 of them on neural differential distinguishers. It identifies comparability, scaling and explainability as open problems [35].
2. **Learned solvers for structured mathematical problems.** The main example is the SALSA family of transformer attacks on LWE with small or sparse secrets [27], [28], [36], [37], [38], together with transformer distinguishers for code-based structures [39].
3. **Agentic LLM systems.** These read specifications, form hypotheses, write SageMath or Python, run experiments and refine their approach [1], [4], [5]. They are now paired with formal-methods tooling, though with limited reliability [14], [40].

Alongside these sits **ML-based side-channel analysis**: profiled and non-profiled deep-learning attacks on traces [41], [42]. It is the most mature line of work against *implementations* of PQC.

2.3 Post-quantum cryptography

Table 2. Post-quantum algorithms relevant to this study (status as of 30 September 2026).

Algorithm	Family	Role	Standard / status	Hard problem	Principal AI-related attack surface observed
ML-KEM	Module lattice	KEM	FIPS 203 (final, 2024) [7]	Module-LWE	Implementation side channels [29], [43], [44]; implementation bugs [30], [31]. No design-level AI result.
ML-DSA	Module lattice	Signature	FIPS 204 (final, 2024) [8]	Module-LWE / SIS	Implementation side channels incl. rejection-sampling leakage [45]. No design-level AI result.
SLH-DSA	Hash-based	Signature	FIPS 205 (final, 2024) [9]	Hash function security	No AI-specific design-level result located.
FN-DSA (Falcon)	NTRU lattice	Signature	FIPS 206 in development; initial public draft pending as of the most recent NIST update we located (Apr. 2026) [46]	NTRU / SIS over NTRU lattices	HAWK attack reported not to transfer [1], [6].
HQC	Code-based (quasi-cyclic)	KEM	Selected Mar. 2025; standard in development [10]	QC syndrome decoding	DL IND-CPA testing found no significant advantage [32].
Classic McEliece	Code-based (Goppa)	KEM	Not selected by NIST; used in some other contexts	Goppa-code indistinguishability, decoding	Transformer distinguisher on Goppa codes in certain settings [39] (not full parameters).
BIKE	Code-based (QC-MDPC)	KEM	Not selected in round 4 [10]	QC-MDPC decoding	First reported ML distinguishers for MDPC/QC-MDPC [39].
HAWK [47]	Module-LIP lattice	Signature	Withdrawn , Jul. 2026 [2], [3]	Module lattice isomorphism	AI-assisted design-level attack [1].

[Fact] NIST SP 800-227 gives requirements for using KEMs, including hybrid combination [48]. NIST CSWP 39 defines cryptographic agility as the capability to replace and adapt algorithms across protocols, software, hardware and infrastructure while preserving security and operations [49]. The NSA’s CNSA 2.0 sets national-security-system timelines in the United States [50].

2.4 Digital forensics

Digital forensics needs *identification, collection, acquisition and preservation* processes that maintain integrity and a documented chain of custody [16], and it needs methods integrated with incident response [17]. In an expert-evidence regime of the Daubert type, the court considers testability, peer review, known or potential error rates, standards and general acceptance [51]. Indian law requires certification for the admissibility of electronic records (Bharatiya Sakshya Adhiniyam 2023, s. 63) [52]. A cryptanalytic conclusion presented as evidence (“this archive was encrypted with a key derivable by attack X”) is an *expert inference* and is subject to all of these requirements.

2.5 Harvest-now, decrypt-later

Mosca’s inequality states that if $x + y > z$, data is at risk. Here x is the confidentiality lifetime of the data, y is the migration time and z is the time until a CRQC exists [12]. Kagai et al. treat HNDL as a temporal risk that depends on data longevity and migration readiness [13]. Blanco-Romero et al. examine the practical feasibility of large-scale harvesting [53]. **[Proposed]** For forensics we add a distinction that the decision-oriented literature mostly leaves implicit. For an artifact that has *already been captured*, later migration does not reduce exposure at all; it only changes the risk to *future* captures. Section 8.3 formalizes this point.

3. Related Work

We group prior work into nine research families. For each, we say what it establishes and what it leaves open for this paper. Section 4 analyses the attack literature in detail, and Appendix B gives the full novelty audit.

(R1) Neural cryptanalysis of symmetric primitives. Gohr’s neural distinguisher for round-reduced Speck32/64 improved on the best classical key recovery for 11 rounds [26]. Benamira et al. showed that such networks partly learn differential–linear features that classical analysis can explain [34]. The six-year survey by Gerault et al. reviews 238 publications, systematically analyses 71 papers on neural differential distinguishers across many round-reduced primitives, and covers explainability and key-recovery methods. It finds that comparability and scaling remain unsolved [35]. In our reading of the work reviewed, reinforcement-learning and evolutionary search act mainly as optimizers *inside* classical frameworks (for example, choosing input differences or characteristics) rather than producing stand-alone breaks. **Gap for this paper:** results are reported as accuracy curves on reduced rounds. There is no standard way to record what was claimed, at what scope, and how it was checked.

(R2) ML attacks on LWE. The Salsa family [27], [28], [36], [37], [38], the TAPAS datasets [54] and the LWE benchmarking study [55] show that transformers can recover *sparse or small* secrets under specific (n , q , h) settings. Section 4.2 gives the details. **Gap:** these attacks target non-standard secret distributions, and their success is reported per configuration. No process links “attack succeeded on configuration X” to “risk to deployed parameter set Y”.

(R3) ML for code-based cryptography. DeepDistinguisher separates Goppa codes from random codes in certain settings and reports the first ML distinguishers for MDPC and QC-MDPC codes [39]. Calderon et al. treat the IND-CPA game as a classification problem and find no significant advantage against ML-KEM, BIKE, HQC or their hybrid combinations [32]. **Gap:** a negative empirical result is evidence of a different kind from a proof. Current reporting does not keep the two apart.

(R4) ML side-channel analysis of PQC. This line of work is the most mature. Deep-learning message recovery succeeds against Kyber masked up to fifth order on Cortex-M4 [29]. A neural multi-valued plaintext-checking oracle cut trace counts by up to 87% for lattice KEMs [56]. A blind ML attack on Kyber needs no ciphertext knowledge [43]. A CNN classifies NTT coefficient pairs at about 96–97% accuracy on an FPGA accelerator [44]. A public-template attack recovers ML-DSA keys with roughly 300 traces on Cortex-M4 [45]. A 2025 survey covers the area [42]. Portability across devices is a long-known weakness [57]. **Gap:** these results are tied to one device and one implementation. A forensic or migration audience needs a statement of transferability that the literature rarely provides.

(R5) LLM and agentic cryptanalysis. This family comprises CryptanalysisBench [4], the AI-grinding workflow [5] and Anthropic’s HAWK and AES reports [1]. CryptanalysisBench already uses *executable*

security games to decide what counts as a break [4]. AI Grinding already insists on “an exact witness and a discriminating control” for every result [5]. **Gap:** both are research-evaluation designs. Neither defines (i) evidence objects that persist and can be audited outside the research setting, (ii) independence requirements for reproduction, (iii) a scope label that prevents a reduced-instance result from being reported as a full break, or (iv) a link to asset risk or migration decisions.

(R6) LLMs combined with formal methods. CrypFormBench (700 instances, 677 schemes, 7 formal languages, 160 properties) finds that LLMs do well at interpreting and completing formal models but poorly at generating, transforming or correcting them; the best score was about 49 out of 100 [14]. CryptoFormalEval couples LLMs with Tamarin to find protocol flaws [40], [58]. **Gap:** formal verification is used *to analyse protocols*. It is not used *to check the AI’s own cryptanalytic claims*.

(R7) AI-assisted PQC migration and cryptographic inventory. Relevant work includes LLM-assisted quantum-safe code auditing [59], static detection of ML-KEM and ML-DSA in stripped binaries for forensics and migration assurance (Kestrel: 128/128 recall with no false positives in its evaluation; 12 previously undocumented ML-KEM-using programs found on a production Linux system) [60], the Crypto-Agility Readiness Score [61], CycloneDX CBOMs [62] and NIST’s agility guidance [49]. **Gap:** inventories record *what is deployed*. They do not record *what has been claimed against it, at what assurance level*.

(R8) Post-quantum digital forensics and evidence integrity. Relevant work includes the Q-Audit Integrity, Q-Non-Equivocation and Q-Binding security notions for post-quantum audit evidence, with hybrid re-signing and Merkle-root anchoring [23]; QFRA for hybrid classical–quantum networks [63]; the Evidence Record Syntax [64] and a post-quantum, algorithm-agile ERS profile under discussion at the IETF [65]; and tamper-evident logs [19], [20]. **Gap:** these protect *evidence integrity*. They do not address *cryptanalytic claims as evidence*.

(R9) Assurance of AI outputs and agent actions. Relevant ideas include proof-carrying code [66], cryptographic certificates that an AI agent’s action satisfies a formal policy [67], capability-bound certificates and reproducibility commitments for agent tool use [68], and software supply-chain provenance [21], [22]. **Gap:** these mechanisms are generic. None of them handles the semantics of a cryptanalytic claim, such as scope, attack model, or the difference between a statistical and an existential claim.

4. State-of-the-Art Analysis

4.1 Classical and neural cryptanalysis of symmetric primitives

[Published] Neural distinguishers are well established for *round-reduced* lightweight ciphers [26], [35]. They are learned estimators of a distinguishing advantage under chosen-plaintext access. Their limits are well documented: large data requirements, results tied to particular input differences, and difficulty scaling to more rounds or larger states [35]. AI-assisted results on full AES, full SHA-2 or full SHA-3 would be design-level breaks. We found no peer-reviewed or credible preprint evidence of any such result. Anthropic’s report mentions preliminary, unverified observations on several other primitives (including SHA-1, Salsa20, Serpent and LEA) and states that these remain under investigation [1]. We therefore treat them as unverified hypotheses (assurance level L0 in Section 9).

4.2 AI against lattice cryptography

Table 3 summarizes the main lattice results. All parameter details come from the cited primary sources. “Not reported” means we could not verify the detail in the abstract or summary we consulted. It does not mean the detail is absent from the full paper.

Table 3. AI/ML attacks on lattice problems and schemes.

Work	Target	Parameters and secret distribution	Data / training	Compute (as reported)	Outcome	Full standardized parameters?	Class
SALSA (NeurIPS 2022) [36]	LWE	Small n ; sparse binary secrets	Millions of LWE samples	Not reported here	Recovers secrets in small dimension	No	B

Work	Target	Parameters and secret distribution	Data / training	Compute (as reported)	Outcome	Full standardized parameters?	Class
SalsaPicante (CCS 2023) [27]	LWE	$n \leq 350$, h up to $\approx n/10$ ($h = 60$ at $n = 350$); binary secrets	$4n$ samples, then preprocessing via lattice reduction	Not reported here	Secret recovery; authors state it “does not threaten NIST’s proposed LWE standards”	No	B
SALSA VERDE (NeurIPS 2023) [37]	LWE	Up to $n = 512$; binary, ternary, narrow-Gaussian sparse secrets; smaller q	As above, improved	Reduced vs. predecessors	Extends secret distributions	No	B
Salsa Fresca (TMLR 2025) [28]	LWE	First ML recovery of sparse binary secrets at $n = 1024$ (homomorphic-encryption-relevant setting)	$\approx 10\times$ better sample efficiency via pre-training	$\approx 25\times$ faster preprocessing	Secret recovery in sparse regime	No (sparse binary secrets are not ML-KEM’s distribution)	B
Benchmarking LWE attacks (S&P 2025) [55]	Kyber-like parameters with low-weight secrets	Binomial secrets of Hamming weight 9–11 for Kyber ($\kappa = 2$)	Comparison of SALSA, Cool & Cruel, MitM, uSVP	28–36 h for SALSA / Cool & Cruel; uSVP failed after $> 1,100$ h	Recovers <i>low-weight</i> secrets only	No: standardized parameters are not broken	B
Data repetition and stepwise regression (2026) [38]	LWE	$n = 256$ ($\log_2 q = 12, 20$) and $n = 512$ ($\log_2 q = 28, 41$); recoverable h rises, e.g. $33 \rightarrow 70$ at $n = 256$, $\log_2 q = 20$	400 M synthetic samples	≈ 42 M CPU-hours BKZ; ≈ 5 GPU-days per model	Up to 8 “cruel bits” recovered	No	B
HAWK (Anthropic, 2026) [1], [15]	HAWK signature (module-LIP)	Specified parameter sets; HAWK-256 $2^{64} \rightarrow 2^{38}$ per Anthropic. A secondary analysis reports larger HAWK-512 and HAWK-1024 reductions under several cost models, which we have not verified [15]	n/a (mathematical insight)	≈ 60 h; $\approx$ USD 100k API cost	Faster exponential-time key recovery; “does not run in polynomial time”	Yes for HAWK (security margin cut; scheme withdrawn)	A + E

[Fact] No ML or LLM result we reviewed affects the standardized secret distributions or parameter sets of ML-KEM or ML-DSA at design level. A 2026 preprint series that claimed a *quantum* attack on ML-KEM was found by independent analysts to contain fatal mathematical errors; for example, the attacked ideal is independent of the secret [69]. This episode shows why scope and verification labels are needed.

4.3 AI against code-based cryptography

[Published] DeepDistinguisher (SAC 2025) is a transformer that distinguishes Goppa codes from random linear codes. It “outperforms traditional attacks in distinguishing Goppa codes in certain settings” and gives the first distinguishing results for MDPC and QC-MDPC codes [39]. The summary we verified does not establish distinguishing at full Classic McEliece parameters. A structural distinguisher would also be only one step towards a decoding or key-recovery attack. [Published] Deep-learning IND-CPA testing of BIKE, HQC and hybrid combinations found no statistically significant advantage [32]. [Fact] This is a negative result under a particular learner and data budget. It is not a security proof.

4.4 Hash-based, multivariate, isogeny and symmetric components of PQC

- **Hash-based (SLH-DSA, SHA-2/SHA-3/SHAKE).** We did not locate any AI-derived design-level result. Neural distinguishers on reduced-round hash components appear in the survey literature [35], but nothing scales to the full functions used in FIPS 205 [9]. [Open] An AI-assisted search for parameter-misuse or implementation faults in SLH-DSA, such as fault attacks on the hypertree, is a plausible research target. We do not treat it as an established result.
- **Multivariate.** The decisive break of Rainbow was human and algebraic [25]. AI Grinding reports an *assumption-level* failure in a multivariate MinRank hardening construction, not a complete break [5]. The round-3 additional-signature candidates MAYO, QR-UOV, SNOVA and UOV [3] have no AI-derived design-level result that we could verify.
- **Isogeny.** The SIDH break was human [24]. We found no AI-derived result against SQIsign [3].
- **Symmetric components inside PQC.** These are SHAKE, AES (for example in AES-based KEM variants) and the XOFs used in ML-KEM and ML-DSA. The only AI-assisted results are reduced-round (the 7-round AES result) [1], [6].

4.5 AI side-channel cryptanalysis of PQC implementations

Table 4. Representative ML side-channel results on PQC implementations.

Work	Target / platform	Leakage / model	Data	Reported result	Profiled ?	Generalization evidence
Dubrova et al. [29]	Masked Kyber (orders ≤ 5), ARM Cortex-M4	Power; DL with “recursive learning” (weights copied from lower-order models)	Profiling traces	Message-bit recovery > 99%	Yes	Single platform
Tanaka et al. (TCHES 2023) [56]	Kyber, Saber, FrodoKEM, NTRU Prime, SIKE	Multi-class NN plaintext-checking oracle on FO re-encryption	Attack traces	Up to 87% fewer traces for lattice KEMs at 99.9999% success	Yes	Three RO instantiations
Ravi et al. (ICCAD 2023) [43]	Kyber (pqm4), Cortex-M4	HW classifier on pointwise multiplication; blind (no ciphertext)	Few hundred to few thousand traces	Full key recovery, given an accurate HW classifier	Yes	Single platform
Chinbat et al. (2025) [44]	Kyber NTT accelerator, FPGA (CW305)	12 models (MLP, CNN, RNN)	200,000 traces	Best CNN 96.64% (Kyber768/1024), 95.71% (Kyber512) pairwise-coefficient accuracy	Yes	Unprotected design; authors note masking may change results
Zhao et al. (DATE 2026) [45]	ML-DSA-44 (and other levels), Cortex-M4	Public-template, non-profiled; rejected and accepted signing trials	Challenge recovered with 96 traces; key ≈ 300 traces	Key recovery	No clone device	Single platform
Rezaeizade et al. (USENIX Sec 2025) [70]	General (blind DL-SCA)	DL without plaintext/ciphertext knowledge	—	Blind DL key recovery	No	—

[Published] Three implementation-class results show that PQC software is attackable through ordinary engineering defects: the KyberSlash timing leak from secret-dependent division [30]; the incomplete Fujisaki–Okamoto ciphertext comparison in wolfSSL’s SIMD ML-KEM paths, which allowed key recovery from a few hundred chosen ciphertexts and was fixed in wolfSSL 5.9.2 [31]; and DL attacks on masked code [29]. [Fact] None of these is a break of FIPS 203 or FIPS 204.

Model-family comparison. Across the surveyed work [41], [42], CNNs dominate profiled attacks on misaligned traces because they tolerate shifts. MLPs are competitive on well-aligned points of interest. Recurrent models (LSTM) appear less often and offer no consistent advantage. Transformers are emerging but have not shown a consistent margin over CNNs in the PQC results we could verify. Classical template attacks and CPA remain strong baselines when the leakage model is known; Zhao et al. use CPA-style statistics within

a public-template approach [45]. **[Limitation]** Because datasets and targets differ, no controlled head-to-head comparison across these families on PQC exists. This motivates experiment E6 (Section 11).

Portability. Profiled models trained on one device often degrade on another identical-looking device, because of process variation, clock jitter and measurement setup. Compiler and implementation changes add further shift [57]. Almost every PQC result in Table 4 is single-device. **[Hypothesis H5]** For a forensic or migration audience, a single-device result should be reported as implementation- and device-scoped until cross-device transfer is measured.

4.6 LLM and agentic cryptanalysis: what did the AI actually contribute?

Table 5 applies the eight questions of our analysis protocol to each 2026 claim.

Table 5. Critical analysis of 2026 AI-generated cryptanalytic claims.

Question	HAWK [1], [6], [15]	7-round AES-128 [1], [6]	CryptanalysisBench novel findings [4]	AI Grinding [5]
1. What did the AI contribute?	Found a non-trivial automorphism (a Galois involution, per secondary analysis [15]) in HAWK’s lattice. Prior theory had shown that such an automorphism would help, but none was known for HAWK	A fingerprinting technique that reduces work by a factor of 256; overall time $\approx 2^{89}$	Agent-generated attacks that win executable security games	Autonomous generate–test–refine; returns witnesses, controls and code
2. Was the mathematics new?	Low novelty: it extends known tools [6]	Modest constant-factor improvement on 2012–2013 work [6]	A SpoC key recovery and a KINDI proof error, reported as previously unknown	Exact failures in eight constructions; two further assumption-level failures
3. Recombination of existing techniques?	Yes: known lattice-isomorphism and reduction methods, applied where no one had looked	Yes	Partly; the benchmark separates recall from discovery	Largely yes (“public algebraic operations that expose hidden relations”)
4. Independently verified?	End-to-end verification pipeline and demonstration code from the reporting team; acknowledged by the HAWK team through withdrawal [2]	Validated by the reporting team over several hundred human hours; Green notes it is unclear how “real” the speed-up is without a full implementation [6]	A break counts only if it wins the formal game; independence across models (two models found SpoC independently)	Every result has “an exact witness and a discriminating control”
5. Full-strength target?	Yes for HAWK’s specified parameters. The <i>practical</i> run was on a weakened challenge instance [6]	No (7 of 10 rounds)	Some Tier-2 schemes broken at full strength (8.9% for the best model)	Yes for the eight published constructions
6. Security level affected?	HAWK-256: $2^{64} \rightarrow 2^{38}$. Larger sets “remain impractical to attack”	None for full AES	Lesser-studied schemes	Specific constructions
7. Transfers to related schemes?	Reported not to affect Falcon, ML-DSA or other lattice schemes [1], [6]	No	Scheme-specific	Pattern-level lessons, construction-specific breaks
8. Assumptions needed?	Exponential-time cost model; Core-SVP-style estimates	Chosen plaintext; 2^{105} chosen plaintexts	Oracle access as defined by each game	As stated per construction
Our assurance level (Section 9)	L1 + L2 (executable; acknowledged by the designers)	L1 on paper ; L3 not established	L1 (game-verified) per finding	L1 per witness; L2 pending external replication

4.7 AI failure modes in cryptanalysis

Table 6 lists 26 failure modes. “Existing mitigation” names what the literature already does. “Remaining gap” is the part our framework targets or leaves open.

Table 6. Taxonomy of AI cryptanalysis failure modes.

#	Failure mode	Cause	Consequence	Existing mitigation	Remaining gap (→ our mechanism)
1	Hallucinated attack	Fluent generation without checking	Invalid attack enters reports	Executable games [4]; witnesses [5]	No persistent claim record → evidence object C (§9)
2	False cryptanalytic positive	Weak test, leakage of the answer, bad statistics	Unreproducible “break”	Controls [5]	No pre-registered statistical gate → Table 9
3	False negative	Limited search, prompt sensitivity	False assurance	Multiple models [4]	Negative results not scoped → “negative” claim form
4	Training-data leakage (recall presented as discovery)	Attack in pre-training corpus	Over-stated novelty	Tier-1 vs. Tier-2 separation [4]	No novelty field in claims → H.novelty
5	Benchmark contamination	Public tasks reused	Inflated benchmark scores	Held-out or unbroken tasks [4]	→ Sealed, dated hallucination traps (§12)
6	Overfitting	Small or idiosyncratic data	Collapse on fresh data	Validation splits	→ Independent-oracle re-test
7	Data hunger	ML-LWE, DL-SCA	Unrealistic attack models	Pre-training [28]	Data budget not in claims → H.resources
8	Unrealistic assumptions	Chosen-plaintext, clone devices, sparse secrets	Inapplicable to practice	Author caveats [27]	→ Mandatory attack-model field
9	Reduced-parameter bias	Feasibility on toy instances	Reduced result reported as full break	Careful reporting [6]	→ Scope-Inflation Rate metric
10	Model drift	API model updates	Results not reproducible later	Model version logging	→ P.model hash/version, sealed replay
11	Proprietary-model reproducibility	Closed weights	Cannot re-run	Release transcripts [1]	→ Reproduction must not depend on the generating model
12	Prompt sensitivity	Different phrasing gives different output	Unstable findings	Multiple runs	→ Record prompt, seed and sampling parameters
13	Adversarial input manipulation	Crafted binaries or specs	Mis-identification	—	→ Security analysis (§17)
14	Training-data poisoning	Compromised traces or corpora	Backdoored attack models	Data hygiene	→ Dataset hashes in P
15	Side-channel distribution shift	New device or setup	Model fails silently	Domain adaptation [57]	→ Cross-device metric CDG
16	Cross-device / compiler generalization failure	Process variation, compiler flags	Over-generalized claims	Rarely measured	→ Scope: device/compiler fields
17	Implementation confusion	Bug mistaken for a design flaw	“ML-KEM broken” headlines	Expert review	→ Class field (A–E) is mandatory
18	No mathematical proof	Heuristic complexity claims	Speed-ups that are not real [6]	Lean / formal proof (rare)	→ L3 for complexity claims
19	No independent replication	Same model, same team	Correlated errors [15]	Ad hoc	→ Independence relation (§9.3)
20	Computational cost	Large BKZ/DL runs [38]	Unverifiable in practice	—	→ Verification Cost metric
21	Energy cost	GPU training	Sustainability, access inequity	—	[Open] report energy in P

#	Failure mode	Cause	Consequence	Existing mitigation	Remaining gap (→ our mechanism)
22	Forensic admissibility	Opaque method	Exclusion or weight reduction	Expert testimony	Jurisdiction-specific (§18)
23	Evidence contamination	Analysis writes to evidence	Integrity loss	Write-blocking [16]	→ Twin works on hashed copies
24	Privacy leakage	Cloud LLM receives case data	Disclosure	Air-gapping	→ Data-handling policy, local models
25	Chain-of-custody gaps	Unlogged AI steps	Challenge in court	Audit logs [19]	→ Signed append-only log
26	Legal and ethical problems	Dual use, licences, lawful authority	Misuse, liability	Disclosure norms [1]	§19

5. Research Gap

5.1 Evidence-based gap matrix

Table 7. Research-gap matrix.

Existing capability	Existing research	Limitation	Missing integration
AI hypothesis generation	[1], [4], [5]	Research-evaluation setting; claims not persisted with scope or independence	Durable, scoped, graded claim objects
Executable verification of breaks	Security games [4]; witnesses [5]	Covers existential claims well; complexity and statistical claims less well	Verifier chosen by claim form (Table 9)
Formal methods with LLMs	[14], [40]	Formal tools analyse <i>protocols</i> , not the AI’s <i>claims</i>	Formal checking of claim statements (L3)
ML-LWE, ML-code and DL-SCA	[28], [29], [39], [45], [55]	Per-configuration, single-device results	Scope and transfer labels linked to deployed assets
Crypto inventory and migration	[59], [60], [61], [62]	Inventories record what is deployed, not claims against it	Twin edges: asset ↔ claim ↔ assurance level
HNDL modelling	[12], [13], [53]	Decision-level; does not separate already-harvested artifacts	Retrospective vs. prospective risk for forensic triage
PQ evidence integrity	[23], [63], [64], [65]	Protects artifacts, not cryptanalytic inferences	Cryptanalytic claims as preserved evidence
AI-output assurance	[21], [66], [67], [68]	Generic; no cryptanalytic semantics	Scope, attack model, claim form

5.2 The central gap

Taken separately, each research family is active and in some cases mature. What we did not identify, in the literature reviewed through 30 September 2026, is a framework that does all of the following together:

1. treats an AI-generated cryptanalytic claim as a *first-class, scoped, persistently recorded object*;
2. grades it on *separate* assurance levels, with a verifier matched to its logical form and an explicit *independence* requirement for reproduction;
3. preserves it, with its provenance, under *forensic* integrity requirements that survive the post-quantum transition;
4. links it, through a model of the cryptographic estate, to *HNDL exposure and migration decisions*.

5.3 Novelty audit outcome and renaming

The working name “Proof-Carrying AI-Assisted Cryptanalysis” overstates two things.

- **It implies every claim carries a proof.** It does not. Most practical cryptanalytic claims are existential (a key, a forgery, a collision) and are checked by recomputation. Others are statistical and are checked

by hypothesis testing. Only some complexity and proof-error claims can realistically carry machine-checked proofs today.

- **It suggests more novelty than the evidence supports.** The term “proof-carrying” comes from proof-carrying code [66] and is already used for AI agent actions [67]. The idea of “AI output plus executable witness” is already practised by CryptanalysisBench [4] and AI Grinding [5].

We therefore use “**evidence-carrying**” for the overall framework and keep “**proof-carrying**” for the highest assurance tier (L3), where a machine-checked proof really is attached. Appendix B gives the full audit table.

6. Threat Model

6.1 Parties

Adversaries ($\mathcal{A}$). We consider adversaries with combinations of the following:

- (a) classical computation, with a budget measured in core-hours or gate counts;
- (b) frontier AI resources, including agentic LLMs and the ability to train DL-SCA models;
- (c) future quantum resources, modelled as a CRQC that becomes available at an uncertain time Z ;
- (d) passive access to ciphertexts, with harvesting at capture time t_c ;
- (e) access to implementations, including binaries and source;
- (f) side-channel or fault access to devices;
- (g) the ability to submit malicious inputs to the defender’s analysis pipeline, such as crafted binaries, specifications or traces;
- (h) the ability to compromise AI training data, models or prompts.

Defenders ($\mathcal{D}$). The defenders are the forensic investigator, the cryptanalyst, the security engineer and the migration team.

Neutral third parties. These are an independent reproducer, an opposing expert, and a court or regulator.

6.2 Trust assumptions

Table 8. Threat model and trust assumptions.

Component	Trusted?	Rationale	If violated
Evidence acquisition tool	Conditionally: validated, hash-verified build	Standard forensic practice [16]	Use dual-tool acquisition and compare hashes
Cryptographic hash (SHA-256 / SHA3-256)	Yes (collision and second-preimage resistance)	Grover gives only a square-root speed-up for preimages	Algorithm-agile re-hashing is supported (§10)
Log signatures	Yes, for the signature’s lifetime; hybrid classical + ML-DNA or SLH-DNA	PQ evidence notions [23]	Re-sign or anchor before the classical signature becomes unsafe
Verifier (proof checker, reference implementation)	Small trusted base; versioned and hashed	Kernel of a proof assistant [71]; reference code	Use two independent verifiers for L2
Solvers (SAT/SMT, BKZ, estimator)	Not trusted to be complete ; trusted only when they return checkable certificates	Solver bugs exist	Prefer certificate-producing modes; check witnesses independently
Hardware for experiments	Trusted for measurement only, documented	Side-channel results depend on the device	Record device IDs; require cross-device replication
AI models	Never trusted	Hallucination, drift, poisoning (Table 6)	An AI output alone can never raise a claim above L0
Human analyst	Trusted for judgement but logged	Insider risk	Two-person rule for L2/L3 promotion

6.3 Security goals

The framework has five security goals.

- **G1 Soundness of grading.** No claim reaches level $L \geq 1$ unless the verifier required for its claim form at that level accepted it.
- **G2 Scope integrity.** The scope recorded for a claim cannot be wider than the scope that was verified.
- **G3 Provenance integrity.** After a claim is logged, nobody can alter or back-date any component of it without detection.
- **G4 Independence.** An L2 promotion needs a reproduction that is independent on every dimension through which the original error could have propagated.
- **G5 Confidentiality.** Case data does not leave the authorized perimeter. This includes not sending it to third-party AI services unless policy permits.

7. Proposed Framework

[Proposed] The architecture has six layers (Figure 2). The design principle is that **AI generates hypotheses and nothing else; verification, grading and preservation are done by components that do not depend on the AI that produced the claim.**

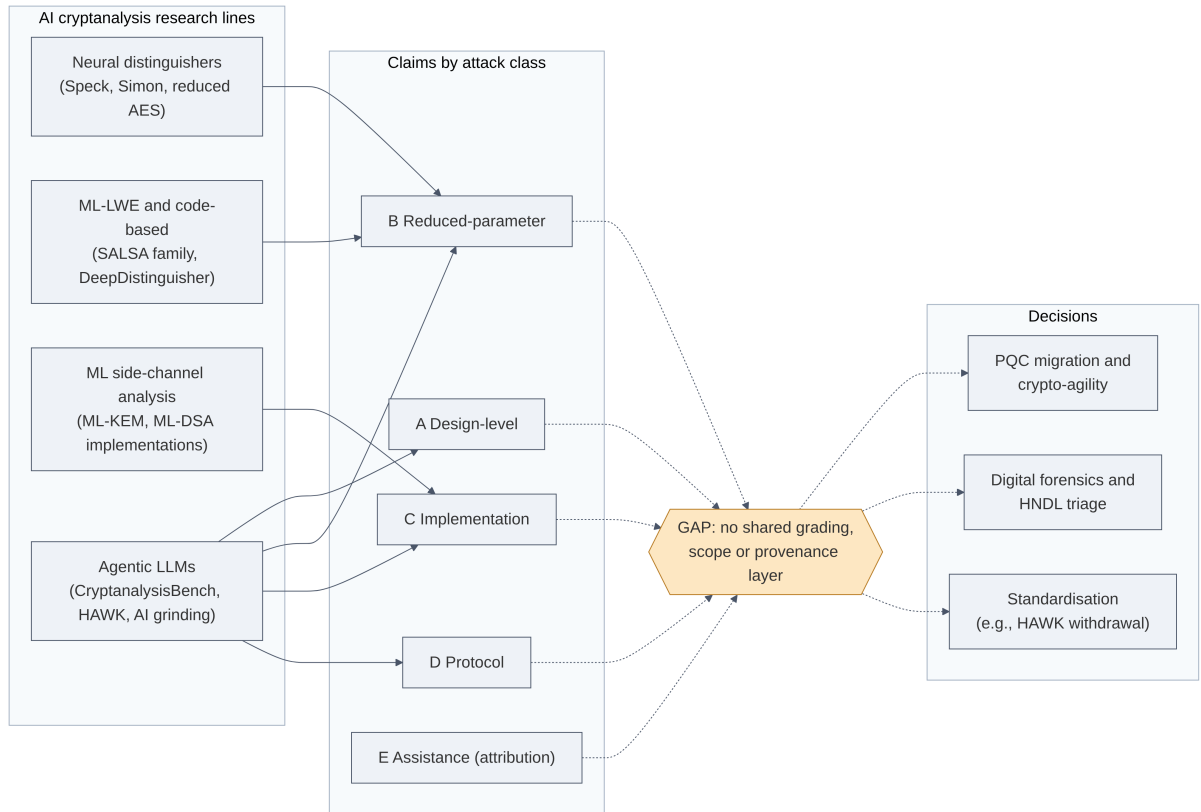


Figure 1. The AI-assisted cryptanalysis ecosystem: four research lines (neural distinguishers, ML lattice and code attacks, ML side-channel analysis and agentic LLMs) feed claims of five attack classes into PQC migration and digital-forensic decisions. Currently there is no shared grading or provenance layer between claims and decisions.

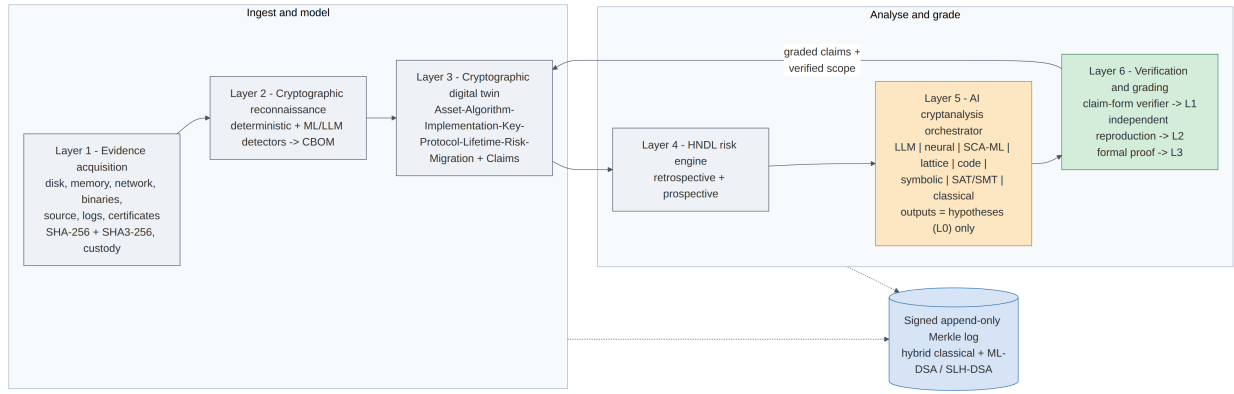


Figure 2. Evidence-carrying AI-assisted cryptanalysis architecture. AI modules (amber) only propose hypotheses. Verification (green) and provenance (blue) are independent of the proposing model.

7.1 Layer 1 — Evidence acquisition

Inputs are disk images, memory dumps, network captures (including TLS handshakes), binaries, source, logs, certificates, key stores, configuration files, and mobile, IoT and cloud artefacts. Each item is acquired with a validated tool, hashed with SHA-256 and SHA3-256 at acquisition, and given a case ID and an evidence ID. Analysis runs only on verified working copies [16], [17]. Acquisition metadata (time, method, operator, tool version and hash) becomes the first entries of the provenance log (§10).

7.2 Layer 2 — Cryptographic reconnaissance

Layer 2 identifies algorithms, parameter sets, implementations, library versions, keys and certificates, protocols and negotiated groups (for example, X25519MLKEM768 versus X25519 alone), and legacy versus PQC usage. It combines deterministic detectors with ML detectors:

- **Deterministic detectors:** constant and NTT-table signatures, which is the approach Kestrel uses for ML-KEM and ML-DSA in stripped binaries [60]; ASN.1 and OID parsing of certificates; handshake parsing; and library version strings.
- **ML or LLM detectors:** classification of obfuscated or inlined code, and contextual triage of source findings [59].

Every detection is itself a claim, of the “identification” form. It is recorded with its detector, its confidence, and a deterministic confirmation where one exists. **[Proposed]** We output reconnaissance results as a CycloneDX CBOM [62], which keeps the results interoperable.

7.3 Layer 3 — Cryptographic digital twin

The twin is a typed, versioned property graph:

Asset → *Algorithm* → *Implementation* → *Key* → *Protocol* → *Data-lifetime* → *Risk* → *Migration-state*

plus **Claim** nodes attached to Algorithm, Implementation or Protocol nodes, with edges labelled by assurance level and scope. The twin lets an investigator answer the following questions:

- *what* is encrypted (asset, sensitivity);
- *how* it is encrypted (algorithm and parameter set);
- *where* it is encrypted (implementation, device, protocol endpoint);
- *how long* confidentiality matters (L_i);
- *whether* the algorithm is quantum-vulnerable, or has open claims against it at a given level;
- *whether* migration has happened, verified behaviourally rather than by the mere presence of a library (§7.7).

“Digital twin” here means a *model of the cryptographic estate*. It does not mean a cycle-accurate hardware simulator. The latter use, which one of the source drafts for this study proposed, is a separate research problem; we leave it to future work (§20).

7.4 Layer 4 — HNDL risk engine

Section 8.3 defines the risk engine formally. It ranks artifacts and flows by retrospective and prospective HNDL risk, and it writes the scores and their parameter provenance back into the twin.

7.5 Layer 5 — AI cryptanalysis orchestrator

The orchestrator dispatches work to specialised modules:

- an **LLM reasoning agent**, which proposes hypotheses and writes analysis code;
- a **neural cryptanalysis module**, for distinguishers and ML-LWE;
- a **side-channel ML module**, for profiled, non-profiled and blind attacks;
- a **lattice module**, which interfaces with reduction tools and the LWE estimator [72];
- a **code-based module**, for decoding and structural distinguishers;
- a **symbolic-mathematics module**, using a computer algebra system such as SageMath;
- **SAT/SMT solvers**;
- **classical tools**, such as MILP and SAT characteristic search [33] and Tamarin for protocols [58].

Every output is a **hypothesis object H at level L0**. The orchestrator may *prioritise* by HNDL risk, but it cannot grade claims.

7.6 Layer 6 — Verification, grading and preservation

This layer selects the verifier from the claim form (Table 9), runs it in a sealed environment, decides promotion under the independence rules (§9.3), and appends signed records to the provenance log (§10).

7.7 Migration verification

[Proposed] We treat migration verification as a set of *negative* and *behavioural* claims, each checked against the twin:

- **(i) No residual classical key establishment on in-scope endpoints.** Checked by active handshake probing: offer classical-only groups and expect refusal where policy requires it.
- **(ii) No unreferenced classical implementations in deployed binaries.** Checked by static detection, for example of RSA, ECDH and ECDSA constants, alongside PQC detection [60].
- **(iii) Downgrade resistance.** Checked by protocol tests, and by Tamarin models where available [58].
- **(iv) Certificate-chain migration.** Checked by parsing chain OIDs.

Figure 6 shows the workflow.

8. Formal Model

8.1 Notation

Let $\mathcal{S}$ be a set of cryptographic schemes, with parameter sets $\mathcal{P}_s$ for each $s \in \mathcal{S}$. An **attack target** is a tuple

$$\tau = (s, p, \iota, \pi, \delta)$$

where $s \in \mathcal{S}$, $p \in \mathcal{P}_s \cup \{\tilde{p}\}$ ($\tilde{p}$ is any non-standard or reduced instance), ι is an implementation identifier (or $\perp$ for the design), π is a protocol context (or $\perp$), and δ is a device or platform descriptor (or $\perp$).

An **attack model** α specifies the oracle access (for example KPA, CPA, CCA, PCO, side-channel, fault), the data budget D and the compute budget T . The **scope** of a claim is

$$\sigma(\tau) = \begin{cases} \text{A (design)} & p \in \mathcal{P}_s, \iota = \perp, \pi = \perp \\ \text{B (reduced)} & p = \tilde{p} \\ \text{C (implementation)} & \iota \neq \perp \\ \text{D (protocol)} & \pi \neq \perp \end{cases}$$

The “assistance” class E is a *provenance* attribute: it records who contributed what, and it is orthogonal to σ .

Definition 1 (scope order). Scopes form a partial order by *strength of the implication for deployed systems*. $A > B$ in the sense that a design-level result implies results for all instances; and $A \not\asymp C$, $C \not\asymp A$, because an implementation break says nothing about the design and vice versa. **Scope inflation** happens when a claim is

reported at a scope σ_r but verified only at a scope $\sigma_v \prec \sigma_r$ with $\sigma_r \neq \sigma_v$. Reporting a B-verified result as an A-claim is the most common case.

8.2 Claim forms

Definition 2 (claim form). Every hypothesis H has exactly one logical form $\phi(H)$:

- **EX — existential witness.** “There is an efficiently checkable object w such that $\text{Check}_\tau(w) = 1$.” Examples: a recovered key, a forged signature, a collision, a plaintext, or a chosen-ciphertext sequence that triggers a decryption failure.
- **ST — statistical advantage.** “Adversary $\mathcal{A}$ achieves advantage $\varepsilon \geq \varepsilon_0$ in game G with D samples.”
- **CX — complexity bound.** “Attack $\mathcal{A}$ recovers the key with cost $\leq 2^c$ in model M .”
- **PD — proof defect.** “Step k of the published security proof is invalid,” supported by a counterexample or a formal refutation.
- **IM — implementation fault.** “Implementation ι on platform δ leaks $f(\text{secret})$ through channel χ ,” or “deviates from the specification at line ℓ .”
- **PR — protocol flaw.** “Protocol π allows downgrade or misuse property ψ .”
- **NG — negative result.** “Within budget (D, T) and learner class $\mathcal{L}$, no advantage above ε_0 was detected.”
- **ID — identification.** “Artifact x uses algorithm s with parameter set p in implementation ι .”

The form determines what can count as evidence. A CX claim backed only by a successful run on a reduced instance has *executable support for a B-scoped EX claim*. It does **not** have evidence for the CX claim at design scope. HAWK illustrates this: the practical demonstration ran on a weakened challenge instance (B, EX), while the security-level statement is a complexity estimate (A, CX) [1], [6].

8.3 HNDL risk model

Setup. For artifact i (a stored ciphertext or a captured session), let:

- $t_{c,i}$ — capture or creation time;
- L_i — required confidentiality lifetime (policy- or law-derived);
- $S_i \in [0, 1]$ — sensitivity, obtained by mapping the data classification through an elicited utility (below);
- A_i — the key-establishment or encryption mechanism protecting i , with parameter set;
- $e_i \in [0, 1]$ — probability that i was exposed to capture (for example, traversed untrusted networks or was stored off-premises);
- Z_A — the random time at which A becomes practically breakable for the relevant adversary class. This is either a CRQC arrival time (for RSA and ECC) or a cryptanalytic hazard (for PQC, informed by the claims registered against A in the twin).

Retrospective risk (already-created artifacts):

$$R_i^{\text{retro}} = S_i \cdot e_i \cdot \Pr[Z_{A_i} \leq t_{c,i} + L_i]$$

Prospective risk (ongoing flows f , before migration completes at time t_m, f):

$$R_f^{\text{pro}} = S_f \cdot e_f \cdot \int_{t_0}^{t_m, f} \lambda_f(t) \Pr[Z_{A_f} \leq t + L_f] dt$$

where $\lambda_f(t)$ is the rate at which sensitive data is generated on flow f .

Properties.

1. Migration status M affects only R^{pro} , through t_m . For harvested artifacts, migration is *irrelevant* to exposure; it matters only for *future* captures. This corrects formulations that use M as a multiplicative discount on all artifacts.
2. The model has **no free weights**. S is an elicited utility, e and λ are measured or estimated frequencies, and $\Pr[Z \leq \cdot]$ is a distribution.
3. Mosca’s inequality [12] is the special case $S = e = 1$ with a degenerate (point-mass) Z , where $R \in \{0, 1\}$ indicates $x + y > z$.

Calibration and validation.

- **Pr[Z ≤ t]** comes from a published or structured expert-elicitation distribution (for example a lognormal fitted to elicited quantiles). It is sensitivity-tested over pessimistic, median and optimistic curves. For PQC mechanisms, the hazard rises when the twin records claims at level ≥ L1 with scope A against A_i. Claims at L0 or with scope B or C do not change the design hazard; scope-C claims instead raise the *implementation* exposure term.
- **S_i** comes from a pairwise-comparison elicitation (for example AHP or Bradley–Terry) with data owners, and is checked for consistency.
- **e_i** is estimated from network topology and logging, for example whether a flow crossed an external link.
- **Validation:** compare rankings with an expert panel’s blinded ranking of a held-out artifact set (Kendall’s τ, NDCG@k); report rank stability under the three Z scenarios; and run a **counterfactual check** on seeded scenarios with known “true” exposure (§12).

If a learned component is used, for example learning-to-rank on artifact metadata, it may only *estimate* *S* or *e*. It does not replace the probabilistic structure. Its errors are reported separately.

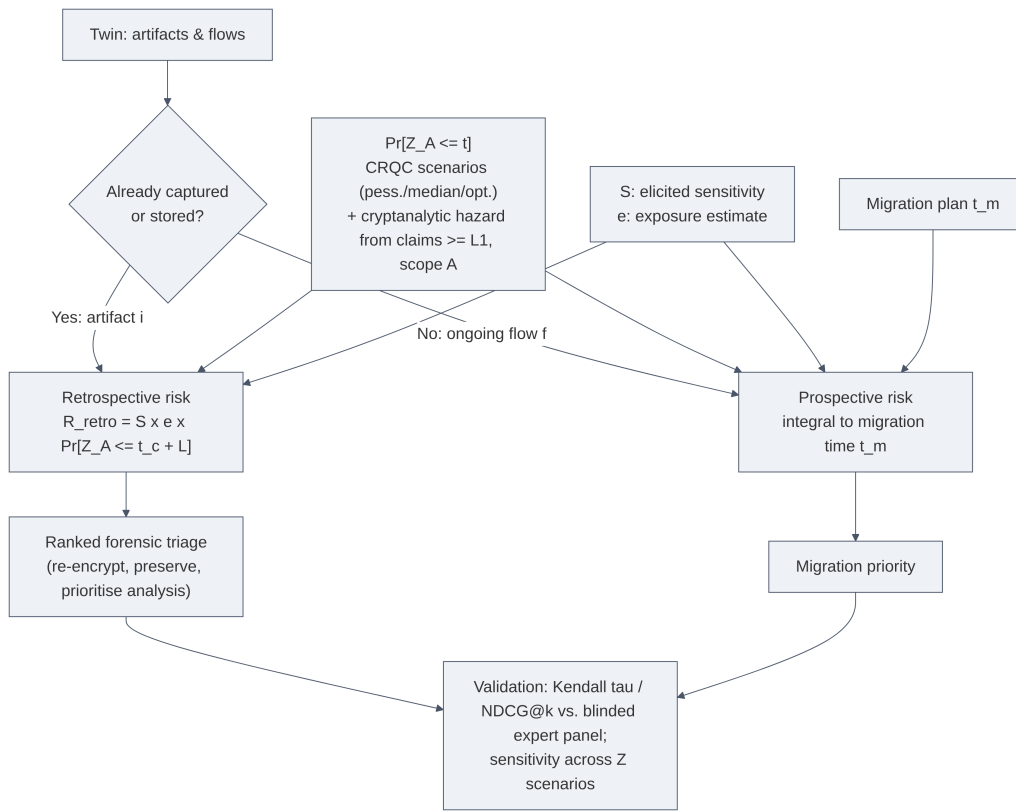


Figure 3. HNDL risk prioritization workflow, separating retrospective risk (already-harvested artifacts, unaffected by migration) from prospective risk (ongoing flows, reduced by migration).

9. Evidence-Carrying (and Proof-Carrying) Cryptanalysis

9.1 The evidence object

Definition 3 (evidence object). A cryptanalytic evidence object is a tuple

$$C = (H, W, E, V, P)$$

- **H — hypothesis.** A structured statement (ϕ , τ , α , σ_{claimed} , statement, novelty, origin), where ϕ is the claim form (Def. 2), τ the target, α the attack model with budgets, σ_{claimed} the claimed scope, *statement* a machine-readable assertion (for example “Check $\tau(w) = 1$ ”, “Adv ≥ 0.05 with $D = 2^{20}$ ”, “cost $\leq 2^{38}$ Core-SVP”), *novelty* $\in \{\text{known, recombination, claimed-new, unknown}\}$ with supporting references, and *origin* recording which agent or human proposed it.

- **W — witness.** Whatever object the claim form calls for. For EX this is the key, forgery, collision or plaintext. For ST it is the trained distinguisher plus its frozen evaluation protocol. For CX it is a derivation, estimator inputs and outputs, and optionally a formal proof script. For PD it is a counterexample or a formal refutation. For IM it is traces, a test harness, and a code location. For PR it is a protocol trace or a Tamarin model. For NG it is the search space explored and the budget spent. For ID it is the matched signature. W can be empty only at L0.
- **E — experimental evidence.** The full experiment record: code (content hash), environment (container digest, library versions), hardware and device IDs, data provenance (dataset hashes, acquisition records), random seeds, raw outputs and run logs.
- **V — verification records.** A *set* of records $v_j = (\text{verifier_id}, \text{verifier_hash}, \text{method}, \text{input_hash}, \text{verdict}, \text{statistics}, \text{independence_vector}, \text{operator}, \text{timestamp})$. There may be several, one per level reached.
- **P — provenance.** Records of how C came to exist and how it changed: model identity and version, prompt and template version, sampling parameters, tool-call transcript hash, analyst identities, chain-of-custody events, and the log position and signatures (§10).

Definition 4 (assurance levels). The levels are separate boolean flags, not a scalar score.

- **L0 — Hypothesis.** H exists. No verifier has run.
- **L1 — Executably supported.** A verifier appropriate to ϕ (Table 9), *different from the generating agent*, accepted (W, E) for the exact τ and α stated in H.
- **L2 — Independently reproduced.** A second verification accepted the claim, independent of the first (Def. 5) on every dimension relevant to ϕ .
- **L3 — Formally verified.** A machine-checked proof of the *statement* exists in a proof assistant or verified tool, against a formal model of τ and α whose adequacy has been reviewed.

These flags are reported together with the **verified scope** σ_v , which can be narrower than σ_{claimed} . A claim can be L3 at scope B and L0 at scope A. Collapsing these into one confidence number would hide precisely the distinction that matters for migration decisions, so we do not do it.

Validity conditions. C is *well-formed* when (i) the hashes in E and V resolve to stored artifacts; (ii) every $v \in V$ references an input hash equal to $H(W \parallel E)$; (iii) $\sigma_v \preceq \sigma_{\text{claimed}}$; and (iv) P contains an unbroken log inclusion proof (§10).

9.2 Verification decision table

Table 9. Verification decision table: which verifier fits which claim.

Claim form	Typical example	L1 verifier	L2 independence requirement	L3 (when feasible)	Common pitfall
EX existential	Recovered ML-KEM key from a buggy implementation; HAWK challenge-instance key	Deterministic recomputation, e.g. decapsulate test vectors, verify a forged signature with the reference implementation	Different reference implementation and machine; witness checked by a second party	Usually unnecessary: the check <i>is</i> the proof	Witness checked against the generating code’s own (buggy) checker
ST statistical	Neural distinguisher advantage; DL-SCA success rate	Pre-registered test on fresh samples from an independent oracle (binomial test, Holm-corrected; α and ϵ_0 fixed beforehand)	Retrain from scratch with different seed, code and preferably model family; new data	Not applicable (empirical)	Test samples generated by the same pipeline; tuning on test data
CX complexity	“ $2^{64} \rightarrow 2^{38}$ ”; 7-round AES cost	Reproduce estimator inputs and outputs (e.g. LWE estimator [72]); run a scaled instance to check the scaling law	Independent derivation by a second analyst or second estimator	Lean/EasyCrypt proof of the cost lemma, given an explicit cost model [71]	Treating a reduced-instance run as evidence of the asymptotic cost
PD proof defect	KINDI CCA-proof error [4]	Concrete counterexample executed against the scheme	Confirmation by the scheme authors or an independent cryptographer	Formalised refutation of the proof step	Misreading the proof’s model

Claim form	Typical example	L1 verifier	L2 independence requirement	L3 (when feasible)	Common pitfall
IM implementation	wolfSSL FO comparison [31]; KyberSlash [30]	Reproduce on a named version and platform; code-level root cause	Second device and/or compiler; maintainer acknowledgement	Verified-implementation diff (e.g. equivalence checking)	Reporting as a design break
PR protocol	Downgrade in hybrid negotiation	Executable protocol trace; Tamarin attack trace [58]	Second model or implementation	Tamarin or ProVerif proof of the fixed protocol	Model abstraction gap
NG negative	“No IND-CPA advantage found” [32]	Power analysis: smallest detectable ϵ at the given D	Different learner family and data	Not applicable	Reported as a security proof
ID identification	“Binary uses ML-KEM-768”	Deterministic signature (constants or OIDs) plus a test-vector run if executable [60]	Second detector	Not applicable	ML detector used alone

9.3 Independence

Definition 5 (independence vector). Each verification v has an independence vector over the dimensions

$\mathcal{J} = \{\text{model family, prompt/template, code base, data source, hardware/device, analyst/organization, verifier implementation}\}$

Two verifications v_1 and v_2 are **ϕ -independent** if they differ on every dimension in the *error-propagation set* $E(\phi)$:

- $E(\text{EX}) = \{\text{verifier implementation, hardware}\}$
- $E(\text{ST}) = \{\text{code base, data source, model family, hardware}\}$
- $E(\text{CX}) = \{\text{analyst/organization, verifier implementation}\}$
- $E(\text{IM}) = \{\text{hardware/device, code base}\}$
- $E(\text{PR}) = \{\text{verifier implementation, analyst/organization}\}$
- $E(\text{NG}) = \{\text{model family, data source}\}$
- $E(\text{ID}) = \{\text{verifier implementation}\}$

Rationale. These sets list the channels through which the original error could plausibly propagate. Using the *same* LLM to “confirm” a statistical claim with the same code shares the code base and the model family, so it fails $E(\text{ST})$. This formalizes the independence problem [15].

9.4 The pipeline

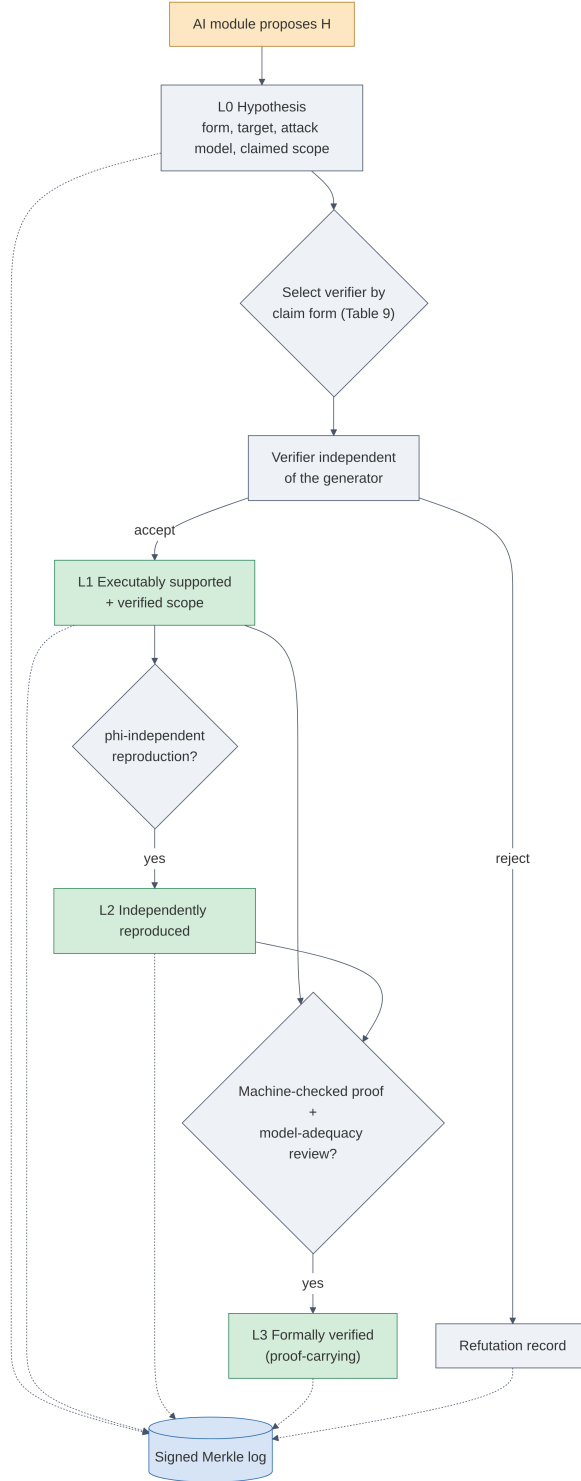


Figure 4. AI hypothesis $\rightarrow$ verification $\rightarrow$ evidence pipeline. Promotion between levels requires a verifier selected by claim form and, for L2, ϕ -independence.

Promotion rules.

- L0 $\rightarrow$ L1 requires an accepting verifier from Table 9 that does not share a code path with the generator.
- L1 $\rightarrow$ L2 requires a ϕ -independent second acceptance.
- Any $\rightarrow$ L3 requires a proof artifact checked by a hashed proof-checker build, *and* a human review of whether the formal model is adequate. A Lean proof about the wrong model is not evidence about the scheme.

- **Demotion.** Any later refuting verification appends a refutation record. The previous records remain in the log, and the claim’s current status is computed from the full record set.

10. Forensic Provenance Architecture

10.1 Record schema

Each log entry is a canonical (deterministically encoded, e.g. CBOR) record. The minimum fields are listed in Table 10. Appendix A gives a JSON-like rendering.

Table 10. Forensic provenance record fields.

Group	Fields
Case	Case ID; legal authority reference; evidence ID; parent evidence ID
Integrity	SHA-256 and SHA3-256 of evidence; acquisition timestamp (trusted time-stamp token); acquisition method and tool hash; evidence source
Actors	Analyst identity (signing key ID); reviewer identity; organization
AI	Model identity, provider, version or weights hash; system prompt or template version; full prompt hash; sampling parameters; random seed; tool-call transcript hash; data-handling mode (local / on-premises / third-party)
Software	Container digest; solver, estimator and proof-checker versions and hashes; library SBOM/CBOM hash [62]
Claim	$H(\phi, \tau, \alpha, \sigma_{\text{claimed}}, \text{statement}, \text{novelty})$; parameters; generated artifacts (hashes)
Verification	V records (verifier hash, method, verdict, statistics, independence vector); independent reproduction references
Conclusion	Current assurance flags (L1/L2/L3), σ_v , narrative conclusion, limitations; reviewer sign-off
Log	Sequence number; previous-entry hash; Merkle root; inclusion proof; signatures (hybrid)

10.2 Is a Merkle tree appropriate?

Yes, for three reasons.

1. Merkle-tree logs give **inclusion proofs** (this entry is in the log) and **consistency proofs** (the log only grew) in $O(\log n)$ [19], [20]. These properties are exactly what is needed to show that no claim was back-dated or removed.
2. A transparency-log design lets **opposing experts or courts** audit the log without access to the underlying case data. They need only hashes.
3. Periodic **signed tree heads** can be anchored in an Evidence Record (RFC 4998) [64]. They can be re-signed or re-hashed under new algorithms when needed, following the post-quantum audit-evidence migration patterns of Kao [23] and the algorithm-agile ERS profile under IETF discussion [65].

Signatures. Log signatures are hybrid: a classical algorithm *and* ML-DSA, or SLH-DSA for long-term anchors. As long as either component is secure, the binding holds [23].

Alternatives considered.

- A plain hash chain is simpler, but it needs $O(n)$ work for audits and has no efficient consistency proofs.
- A blockchain adds a consensus layer that a single forensic laboratory does not need, plus governance and confidentiality costs. QFRA uses a PQC ledger for a different, distributed setting [63].

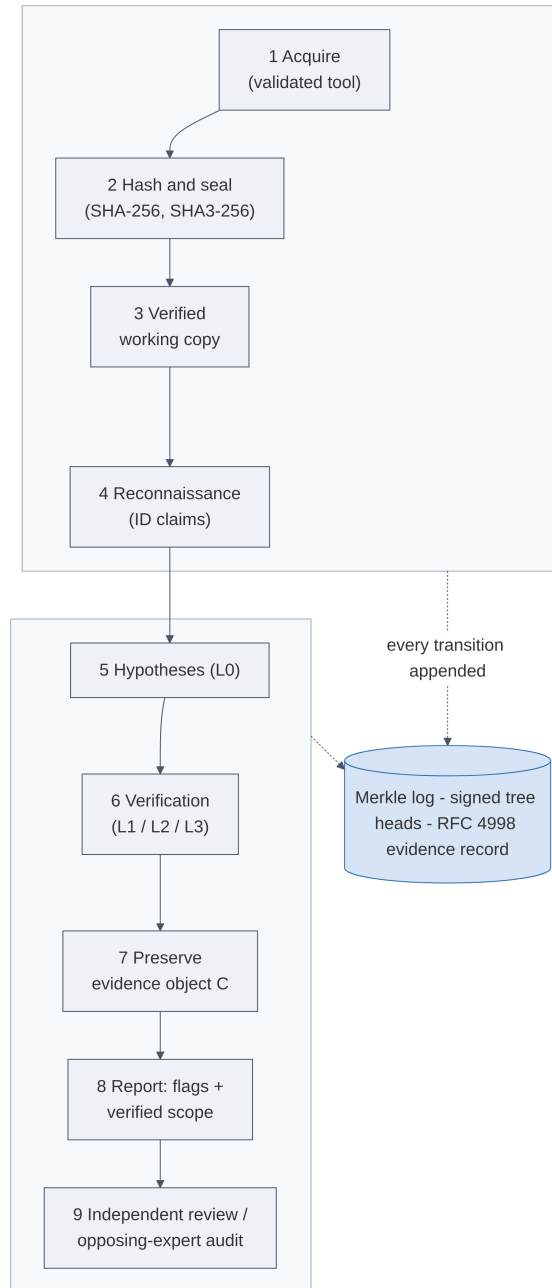


Figure 5. Forensic evidence lifecycle: acquisition, working copy, reconnaissance, hypothesis, verification, preservation and reporting, with each transition appended to the signed Merkle log.

10.3 Sealed replay

For L2 reproduction, E has to be replayable without the generating model. **[Proposed]** We require that the code, data and environment in E are sufficient to re-run the *experiment*. The model transcript in P is kept for audit but is not needed for replay. A closed model can therefore *propose* a claim, but promotion to L2 can never depend on that model being available again. This directly mitigates failure modes 10 and 11 in Table 6.

11. Experimental Methodology

11.1 Research questions (refined)

The research questions were refined from the commissioning list in light of Sections 3–5.

- **RQ1 (identification).** How accurately can a hybrid of deterministic and AI detectors identify algorithm, parameter set, implementation and library version from heterogeneous forensic artifacts?

How much does the AI component add over deterministic detectors such as constant and OID signatures [60]?

- **RQ2 (HNDL prioritization).** Does the two-component risk model (§8.3) rank encrypted artifacts closer to an expert-panel and counterfactual ground truth than (a) recency, (b) algorithm-only and (c) a single multiplicative score that includes migration status?
- **RQ3 (hypothesis utility).** On PQC *implementation* targets with planted and historical faults, and on *reduced-parameter* design targets, what fraction of AI-generated hypotheses reach L1? How does that compare with classical tooling given the same analyst time?
- **RQ4 (reproducibility).** What fraction of L1 claims reach L2 under ϕ -independence? Does this depend on claim form, especially ST versus EX?
- **RQ5 (verification effect).** Does mandatory claim-form-specific verification reduce the false cryptanalytic claim rate and the scope-inflation rate in the *reports* that reach decision-makers?
- **RQ6 (generalization).** How much do DL side-channel attacks on ML-KEM and ML-DSA implementations degrade across devices, compilers, implementations and noise conditions?
- **RQ7 (forensic grade).** Can independent examiners reconstruct, audit and re-run claims from the provenance log alone, and how complete is the provenance?
- **RQ8 (migration verification).** Can behavioural and static checks detect seeded residual classical dependencies and downgrade paths after a declared migration?

11.2 Hypotheses

All hypotheses are directional and will be tested as pre-registered. None has been tested.

- **H1.** AI-assisted workflows reduce analyst effort to reach L1 on implementation-class targets compared with classical-only workflows. We do **not** hypothesize an effect on design-level targets at full parameters.
- **H2.** Claim-form-specific verification (Table 9) reduces the false cryptanalytic claim rate in final reports compared with unverified AI output, and compared with a generic “run the code once” check.
- **H3.** Evidence-carrying records raise the independent reproduction rate and lower the time-to-reproduce compared with conventional write-ups (paper plus code).
- **H4.** Retrospective/prospective HNDL prioritization achieves higher NDCG@k than non-risk-aware ordering and than single-score formulations.
- **H5.** Cross-device transfer significantly degrades DL-SCA attack performance (more traces to rank 1, or lower success rate) on PQC implementations. Domain adaptation reduces but does not remove the degradation.
- **H6.** Without scope labels, AI-generated reports inflate scope (B or C presented as A) at a non-trivial rate. Mandatory scope fields reduce this rate.
- **H0-design (null, expected to hold).** No AI or classical configuration in our study will produce an L1 claim at scope A against ML-KEM, ML-DSA, SLH-DSA or HQC at standardized parameters. This would be a valuable negative result if it holds.

11.3 Baselines

1. **Human-only:** expert analysts with standard tooling, under a time budget.
2. **Classical automated:** MILP/SAT characteristic search [33], the LWE estimator and BKZ [72], CPA and template attacks, static analysers.
3. **AI-only:** LLM agents and neural modules, with output taken at face value.
4. **AI + classical solver:** AI proposes, solvers execute, no grading.
5. **AI + independent verification:** Table 9 verifiers, without provenance or twin.
6. **Full framework:** all six layers.

11.4 Ablation

The full framework is compared against seven ablated versions, each with one component removed: –LLM, –neural module, –verifier (L1 gate), –provenance, –twin, –HNDL prioritization, and –independent reproduction (L2 gate). Section 14.2 lists the expected direction of each effect.

11.5 Experimental matrix

Table 11. Experimental design.

Exp.	Dataset	Attack / task	Baseline(s)	Proposed method	Metrics	Expected observation [Expected]
E1	Forensic corpus F (§12)	Crypto identification	Deterministic-only; LLM-only	Hybrid recon → CBOM	Precision, recall, F1 per field; ID claim L1 rate	Hybrid $\geq$ deterministic on obfuscated/inlined code; LLM-only shows false IDs
E2	Corpus F with seeded lifetimes and sensitivity	HNDL triage	Recency; algorithm-only; single-score	Two-component model	NDCG@k, Kendall τ vs panel; HPP@k	Retrospective term changes top-k vs. single-score
E3	Planted-fault ML-KEM/ML-DSA builds (§12)	Find and confirm implementation on faults	Human-only; classical static/dynamic	Orchestrator + Table 9	Time-to-L1, AER, FCCR	H1 supported on IM claims; FCCR of AI-only > framework
E4	Reduced-parameter LWE (TAPAS [54]) and toy code-based instances	Secret recovery / distinguishing	SALSA-family reference [28]; classical BKZ	Orchestrator with neural + lattice modules	Success rate, L1 rate, CX-claim correctness, SIR	Successes stay at scope B; SIR > 0 when unconstrained
E5	CryptanalysisBench subset [4] + sealed hallucination traps	Agentic cryptanalysis	AI-only	AI + verifier + scope labels	VCR_k, FCCR, SIR, IRR	Traps reveal false positives in AI-only output
E6	Own Cortex-M4 traces (≥ 3 boards, ≥ 2 compilers); ASCAD [41] as sanity check	Profiled / non-profiled DL-SCA on ML-KEM, ML-DSA	Template, CPA; MLP/CNN/LSTM/Transformer	Same, with domain adaptation and CDG reporting	Traces to rank 1, success rate, CDG	H5: significant cross-device degradation
E7	Reproduction track (all L1 claims)	Independent reproduction	Paper + code write-ups	Evidence objects + sealed replay	IRR, time-to-reproduce, EPC	H3
E8	Provenance audit	Third-party reconstruction	Conventional lab notes	Merkle log + records	EPC, audit success, tamper detection	Tampering detected in 100% of seeded cases (by construction; a sanity check)
E9	Migration scenarios with seeded residual classical paths	Migration verification	Library-presence check	Behavioural + static checks	MVA precision/recall	Behavioural checks detect downgrade paths missed by presence checks
E10	Full-parameter ML-KEM, ML-DSA, SLH-DSA, HQC (reference implementations)	Design-level hypothesis generation	Human expert review	Orchestrator	Number of L0 / L1 claims at scope A; refutation reasons	H0-design: no L1 at scope A; every L0 is refuted or stays L0
E11	Full system	Ablation	Full framework	Each component removed	All metrics	Verifier and L2 gate drive FCCR/SIR; twin and HNDL drive E2 metrics

11.6 Statistical methodology

- **Pre-registration.** Hypotheses, primary metrics, $\alpha = 0.05$, minimum effect sizes and stopping rules are registered before running E3–E7 and E10.
- **Paired designs.** The same targets are given to all arms. Binary outcomes are compared with McNemar’s test. Continuous outcomes are compared with Wilcoxon signed-rank tests and reported with Hodges–Lehmann estimates.
- **Intervals.** Rates are reported with Wilson 95% intervals. Ratios and NDCG are reported with BCa bootstrap intervals (10,000 resamples).
- **Multiplicity.** Holm–Bonferroni correction is applied within each experiment family.

- **Human factors** (E3, E7). Mixed-effects models with random effects for analyst and target. Counterbalanced order. Analysts are blinded to arm where feasible. Ethics or IRB approval is obtained before any human-subject study.
- **LLM stochasticity**. At least 5 seeds per configuration. Prompt templates are frozen. Model versions are pinned. Model release dates are recorded relative to task creation dates to bound contamination.
- **Side-channel experiments**. Guessing entropy and success rate are reported as functions of attack traces, averaged over at least 100 random attack subsets.
- **Power**. Before E3, a pilot on 10 targets estimates variance, and target counts are then set for 80% power at the registered minimum effect.

11.7 Mandatory reporting template for design-level targets (Part-19 compliance)

Every claim in E4, E5 and E10 must report the following fields: parameter set; NIST security category; implementation and version; attack model and oracle; data requirement; compute requirement (core-hours and/or Core-SVP or gate-count model); success probability with interval; number of trials; scope (A/B/C/D) and form (ϕ); and assurance flags. Negative results (NG) are reported with the smallest detectable effect size.

12. Benchmark and Dataset Design

[Proposed] We call the benchmark **EVC-Bench**: *Evidence-carrying Verification of Cryptanalytic claims*. It has six tracks. Each track is marked as real-world, synthetic, simulated or benchmark data.

1. **Track K — Known breaks (benchmark)**: a licensed subset of CryptanalysisBench Tier 1 [4], used to calibrate recall-type capability.
2. **Track R — Reduced-parameter lattice and code instances (synthetic)**: TAPAS LWE datasets [54] and generated toy Goppa and QC-MDPC instances [39], with ground-truth secrets.
3. **Track I — Planted implementation faults (synthetic, derived from real code)**. These are ML-KEM and ML-DSA reference and optimized implementations with re-introduced historical fault *patterns*: secret-dependent division timing (KyberSlash-like) [30], incomplete FO comparison [31], and non-constant-time rejection sampling. They are built in an isolated laboratory, with ground-truth fault locations. Fault-free control builds are included. **No new zero-day exploit code is released.**
4. **Track T — Hallucination traps (synthetic)**. These are sealed, dated tasks that look attackable but have no attack within the stated budget, for example full-parameter instances and correct implementations. Any L0 claim of a break is, by construction, a false positive candidate. Trap specifications are kept private until evaluation to limit contamination.
5. **Track S — Side-channel traces (real laboratory data)**. ChipWhisperer-class captures of pqm4 ML-KEM-768 and ML-DSA-44 on at least 3 Cortex-M4 boards, with at least 2 compilers and optimization levels and two masking configurations. ASCAD [41] is used as a public AES sanity baseline. Traces are released together with acquisition provenance.
6. **Track F — Forensic corpus (simulated)**. Synthetic enterprise disk images, memory images, PCAPs (TLS 1.3 with X25519, X25519MLKEM768 and ML-KEM-only groups), certificate stores and binaries. Each has a ground-truth CBOM, seeded data lifetimes and sensitivities, and seeded residual classical paths for migration verification. No real personal data is included.

Data governance. Every dataset is versioned and hashed in the provenance log. Licences are recorded. Access to Track I builds is restricted to vetted researchers.

13. Evaluation Metrics

Table 12. Metrics, definitions and weaknesses.

Metric	Definition	Weakness
VCR_k (Verified Claim Rate at level k ; generalizes PVCR)	$\# \text{claims reaching } \geq L_k / \# \text{claims generated (} k \in \{1, 2, 3\})$	Rewards generating few, safe claims; depends on target difficulty; must be reported alongside the absolute number of verified claims
FCCR (False Cryptanalytic Claim Rate)	$\# \text{claims refuted (or trap-triggered)} / \# \text{claims assessed}$	Needs ground truth; unrefuted $\neq$ true; measured on report contents, not raw generations
SIR (Scope-Inflation Rate) [new]	$\# \text{claims with } \sigma_{\text{claimed}} > \sigma_v / \# \text{claims with } \sigma_v \text{ defined}$	Requires careful scope annotation; inter-annotator agreement (Cohen’s κ) must be reported

Metric	Definition	Weakness
IRR (Independent Reproduction Rate)	#L1 claims reaching L2 under ϕ -independence / #L1 claims	Depends on reproducer skill and resources; slow
EPC (Evidence Provenance Completeness)	Fraction of Table 10 mandatory fields present and resolvable (hashes resolve to artifacts)	Completeness $\neq$ correctness; fields can be filled with wrong data
CTR (Cryptanalytic Time Reduction)	$1 - T_{\text{framework}} / T_{\text{baseline}}$ to reach L1 on the same target	High variance; only meaningful for tasks both arms solve; survivorship bias
AER (Analyst Effort Reduction)	$1 - (\text{analyst-hours}_{\text{framework}} / \text{analyst-hours}_{\text{baseline}})$	Self-reporting bias; learning effects; needs a controlled study
CDG (Cross-Device Generalization)	$N_{\text{traces}}(\text{rank } 1, \text{target device}) / N_{\text{traces}}(\text{rank } 1, \text{profiling device})$, or ΔSR at fixed N	Undefined if the attack fails; device sample size is small
ATT (Attack Transferability)	Fraction of verified attacks that remain L1 on a different implementation or compiler	Depends on the choice of implementations
VC (Verification Cost)	Compute and human hours to reach each level, per claim	Hard to compare across claim forms; L3 cost is dominated by expertise
HPP@k (HNDL Prioritization Precision)	Precision@k (and NDCG@k) of the top-k ranked artifacts vs. ground-truth high-exposure set	Ground truth is itself modelled; sensitive to Z scenario
MVA (Migration Verification Accuracy)	Precision/recall of detected residual classical dependencies and downgrade paths vs. seeded ground truth	Seeded faults may be easier than real ones

Why VCR (PVCR) alone is not enough. A system that proposes only trivial, known attacks scores well on VCR while contributing nothing. A system that proposes bold ideas scores poorly even if one of them is important. We therefore always report VCR_k together with the *absolute* number and novelty class of verified claims, FCCR, SIR and VC.

14. Expected Results and Experimental Validation Plan

No experiments have been run for this manuscript. This section contains no results. It states expected *directions* (labelled [Expected]), the evidence that would falsify them, and the order in which experiments should be run.

14.1 Expected observations and falsifiers

- **[Expected]** E10 yields no L1 claim at scope A against standardized parameters. *Falsifier:* any EX witness that passes Table 9 L1 at a standardized parameter set. This would need immediate coordinated disclosure (§19).
- **[Expected]** AI-only output (baseline 3) has a substantially higher FCCR and SIR than the full framework, because Track T traps and scope labels catch errors that otherwise reach reports. *Falsifier:* no significant difference after Holm correction.
- **[Expected]** For ST claims, IRR is lower than for EX claims, because statistical findings are more sensitive to data and code shared with the original run. *Falsifier:* $\text{IRR}(\text{ST}) \geq \text{IRR}(\text{EX})$.
- **[Expected]** $\text{CDG} > 1$ (degradation) for profiled DL-SCA on PQC targets. Domain adaptation lowers it but it stays above 1. *Falsifier:* $\text{CDG} \approx 1$ across boards and compilers.
- **[Expected]** The retrospective/prospective split changes the top-k triage ranking relative to single-score models whenever migrated and unmigrated systems hold old, high-sensitivity captures. *Falsifier:* rank correlation ≈ 1 between the models across all seeded scenarios.

14.2 Expected ablation effects

Table 13. Ablation study design and expected effects.

Removed component	Primary metrics affected	Expected effect [Expected]	Interpretation if no effect
LLM agent	Number of L1 claims, CTR	Fewer hypotheses on IM/PR targets	LLM adds little beyond classical tooling for these targets

Removed component	Primary metrics affected	Expected effect [Expected]	Interpretation if no effect
Neural module	Track R, Track S success	Lower success on reduced LWE and SCA	Classical solvers suffice at tested scales
Verifier (L1 gate)	FCCR, SIR	Large increase	Verification redundant (unlikely; would undermine C1)
Provenance	EPC, IRR, time-to-reproduce	EPC drop; slower reproduction	Conventional write-ups suffice (would weaken C2)
Digital twin	HPP@k, MVA	Lower triage and migration accuracy	Flat inventories suffice
HNDL prioritization	HPP@k, time-to-first-critical	Lower	Ordering does not matter at tested corpus sizes
Independent reproduction (L2 gate)	Share of claims later refuted	More late refutations	L1 is a sufficient bar for these claim forms

14.3 Validation roadmap

1. **Phase 1:** E1, E2 and E8 (no offensive capability needed; validates C2 and C3).
2. **Phase 2:** E3, E4, E5 and E7 (validates C1 and C4).
3. **Phase 3:** E6 (laboratory hardware).
4. **Phase 4:** E9, E10 and E11.

Results are to be published as evidence objects, with the provenance log's signed tree heads, so that reviewers can audit the paper's own claims with the framework.



Figure 6. PQC migration verification workflow: declared migration state in the twin is tested by static detection, active handshake probing, downgrade testing and certificate-chain analysis; each check is an NG or ID claim graded and logged like any other.

15. Discussion

15.1 Evaluating the central proposition

Evidence for. The proposition is supported in three areas:

- **Lesser-studied and candidate designs.** AI produced a verified design-level result against a NIST candidate (HAWK) [1], [2], game-verified breaks of lesser-studied schemes [4], and witness-backed failures in published constructions [5].
- **Implementations.** ML side-channel attacks on PQC implementations are mature and keep improving [29], [45], [56].
- **Speed.** Hypothesis generation has become cheap. The AES case reports a billion output tokens produced autonomously, against hundreds of human hours of validation [1].

Evidence against or limiting. Four observations limit the proposition:

- The decisive historical PQC breaks (SIDH, Rainbow) were human [24], [25].
- No AI result affects ML-KEM, ML-DSA, SLH-DSA or HQC at design level.
- The deep-learning IND-CPA tests found no advantage [32].
- The HAWK insight was, in Green’s assessment, low in mathematical novelty [6]. AI’s current comparative advantage therefore appears to be *breadth and persistence*, not new mathematics.

Our reading. “AI changes the threat landscape” is supported for implementations and for the *long tail* of less-scrutinized designs. It is not supported for the design-level security of the standardized schemes. The policy conclusion — that verification and provenance mechanisms are needed — does not depend on how often AI is

right. It follows from the fact that AI-generated claims now arrive in volume and are hard to check. That conclusion therefore holds even under the conservative reading.

15.2 Implications for PQC mitigation

1. **Diversity matters.** The withdrawal of HAWK after an AI-assisted analysis [2] illustrates why NIST’s portfolio includes schemes built on unrelated assumptions (SLH-DSA, HQC) [9], [10]. It also shows why agility is a design requirement [49].
2. **Implementation assurance is where AI pressure is most concrete.** The recent ML-KEM incidents were implementation faults, not standard flaws [30], [31]. Migration verification should test behaviour, not merely the presence of a library.
3. **Harvested data cannot be fixed by migration.** Forensic triage must rank harvested artifacts by retrospective risk (§8.3).

15.3 Comparison with prior work

Table 14. Proposed framework vs. closest prior work.

Capability	CryptanalysisBench [4]	AI Grinding [5]	CrypForm Bench / CryptoFormalEval [14], [40]	Kestrel / code auditing [59], [60]	PQ audit evidence [23]	This paper
AI hypothesis generation	✓	✓	✓ (formal models)	partial (LLM triage)	—	✓ (orchestrated)
Executable verification	✓ (games)	✓ (witness + control)	✓ (formal tools)	✓ (deterministic)	—	✓ (by claim form)
Claim-form-specific verifier selection	implicit	implicit	—	—	—	✓ (Table 9)
Explicit scope label (A/B/C/D) and SIR	partial (tiers)	partial	—	—	—	✓
Independence-controlled reproduction	partial (multiple models)	—	—	—	—	✓ (Def. 5)
Persistent, signed provenance	—	run records	—	—	✓	✓
Link to assets / HNDL / migration	—	—	—	✓ (inventory)	—	✓ (twin + risk)
Forensic chain of custody	—	—	—	partial (forensic use case)	✓ (audit evidence)	✓
Experimental validation	✓	✓	✓	✓	case study	✗ (planned only)

15.4 Threats to validity

- **Construct validity:** ground truth for “true” HNDL exposure is modelled, not observed.
- **Internal validity:** planted faults may be easier to find than real faults.
- **External validity:** laboratory devices and synthetic corpora may not reflect production estates.
- **Model churn:** frontier models change faster than a study can run, so results describe model *versions*.

16. Limitations

- **[Limitation]** No experiments have been conducted. Every effectiveness claim is a hypothesis.
- **[Limitation]** L3 is expensive and rarely feasible for CX claims about real schemes. Formalizing cost models is itself hard, and LLM help with formal models is unreliable [14].
- **[Limitation]** The independence sets $E(\phi)$ are proposed from first principles. They need empirical validation, for example by measuring correlated failures.
- **[Limitation]** The HNDL model’s usefulness depends on the CRQC-arrival distribution, which is deeply uncertain. We address this with scenarios, not by resolving it.

- **[Limitation]** Several 2026 sources are preprints, blog posts or vendor reports, not peer-reviewed papers [1], [4], [5], [6], [15]. They must be re-checked against their final versions.
- **[Limitation]** We could not access the full text of every cited paper. Some parameter details in Tables 3 and 4 are taken from abstracts and author summaries, and Table 3 marks the details that are missing.
- **[Limitation]** Legal analysis is limited to identifying issues; it is not legal advice.

17. Security Analysis of the Framework

Table 15. Attacks on the framework and mitigations.

Attack	Vector	Impact	Mitigation	Residual risk
Prompt injection	Strings in evidence (binaries, docs, certificates) read by the LLM agent	Agent misreports or exfiltrates data	Treat evidence as data; tool sandbox; no network egress; the LLM cannot grade (G1)	Agent may waste effort; it cannot promote claims
Adversarial examples	Crafted code or traces mislead ML detectors	Wrong identification	ID claims need deterministic confirmation (Table 9)	Inputs that fool both detectors
Data / model poisoning	Corrupted training traces or corpora	Backdoored SCA or ID models	Dataset hashes in P; train on curated data; L2 requires a different data source	Poisoning of a shared public dataset
Evidence poisoning	Suspect plants decoy crypto artifacts	Misdirected triage	Cross-check against acquisition metadata; analyst review	Sophisticated anti-forensics
Malicious ciphertexts / files	Inputs that crash or exploit tools	Tool compromise	Isolated workers; memory-safe parsers; hashed tool builds	Zero-days in tools
Forged provenance	Fabricated log entries	False chain of custody	Signed entries; Merkle consistency proofs; external tree-head anchoring [20], [64]	Signing-key compromise → key rotation, threshold signing
Replay	Re-submission of old verification as new	Inflated levels	Verification bound to input hash and log position; nonces	—
Insider	Analyst promotes a claim improperly	False L2/L3	Two-person rule; independence vector recorded and audited	Collusion
Model extraction	Querying deployed SCA models	Leaks attack capability	Access control; rate limits	—
Sensitive-data leakage	Case data sent to third-party AI	Confidentiality breach (G5)	Local or on-premises models by default; policy field in P	Policy violations
Compromised verifier	Bugged or malicious proof checker or solver	Unsound promotion	Small trusted base; hashed builds; diverse verifiers for L2	Common-mode bugs

18. Digital Forensic and Legal Considerations

- **Chain of custody and integrity.** The log and hashing design (§10) supports the documentation that forensic standards expect [16], [17]. Whether it *satisfies* a particular court’s requirements is a jurisdiction-specific legal question.
- **Reliability and error rates.** Daubert-type inquiries consider testing, peer review, error rates, standards and general acceptance [51]. FCCR, IRR and VC (§13) are designed to produce the error-rate evidence such an inquiry asks for, but only once they are measured. A proposed U.S. rule that would apply expert-reliability standards to machine-generated evidence offered without an expert was returned for revision in June 2026 [18]. **[Requires jurisdiction-specific legal review.]**
- **India.** Electronic records require certification under s. 63 of the Bharatiya Sakshya Adhiniyam 2023 [52]. How a signed provenance log maps onto the certificate requirement **requires review by Indian counsel.**
- **EU.** The AI Act [73] may impose obligations on AI systems used in law enforcement contexts. Whether a given deployment of the orchestrator falls in a regulated category **requires legal review.**

- **Explainability.** At L1, an EX witness is self-explanatory: the key works. ST and CX claims need an expert narrative. The framework records the verification, but it does not replace expert testimony.
- **We make no claim** that any output of this framework is admissible in any jurisdiction.

19. Ethical and Dual-Use Considerations

- **Dual use.** The same orchestrator that helps defenders can help attackers. We restrict release of Track I builds, do not publish new exploit code against deployed software, and follow coordinated disclosure, as practised for HAWK and the wolfSSL issue [1], [31].
- **Lawful authority.** Cryptanalysis of seized material must be authorised. The case record (Table 10) includes a reference to the legal authority.
- **Privacy.** The default is local models and no third-party processing of case data.
- **Model licences.** Some AI providers restrict offensive use, so licences are recorded in P.
- **Credit and accountability.** The E (assistance) attribution in P records the contributions of humans and AI. Humans remain responsible for any conclusion that is reported.
- **Equity and energy.** Verification costs may concentrate capability in well-resourced laboratories. We report VC and energy use to make this visible.

20. Future Work

1. Empirical validation of E1–E11 (§14.3).
2. Measuring correlated failure across models in order to validate the independence sets $E(\phi)$.
3. Formal libraries of cost models (Core-SVP, gate-count) in Lean, to make L3 practical for CX claims.
4. Integration with the post-quantum ERS profile under IETF discussion [65].
5. A device-level side-channel twin: simulated leakage compared with measured leakage, as an anomaly detector.
6. Applying the framework to the remaining additional-signature candidates in a defensive, pre-standardization role [3].
7. Community governance: a shared registry of evidence objects for AI-generated cryptanalytic claims.

21. Conclusion

AI-assisted cryptanalysis produced real results in 2026. These include a design-level reduction of a NIST candidate’s security that led to its withdrawal, game-verified breaks of lesser-studied schemes, and a growing body of ML implementation attacks on PQC. None of these results weakens the standardized ML-KEM, ML-DSA or SLH-DSA designs. The risk to migration and forensics is therefore not that “AI has broken PQC”. It is that AI produces more claims than anyone can check, of uneven validity and often unclear scope.

We proposed an evidence-carrying framework that grades each claim by its logical form, scope and independently reproduced status. It preserves each claim as signed, Merkle-logged evidence, and connects it to HNDL exposure and migration verification through a cryptographic digital twin. The framework is a design and an evaluation protocol, not yet a validated system. Its value depends on the experiments laid out in Section 14.

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Appendix A. Evidence Object Schema (illustrative)

```

EvidenceObject {
  id, case_id, evidence_ids[], log_seq, prev_hash
  H: { form: EX|ST|CX|PD|IM|PR|NG|ID,
      target: {scheme, param_set, nist_category, impl, version,
              platform, protocol},
      attack_model: {oracle, data_budget, compute_budget,
cost_model},
      scope_claimed: A|B|C|D,
      statement, novelty: known|recombination|claimed-new|unknown,
      novelty_refs[], origin: {agent_id|analyst_id} }
  W: { type, artifact_hashes[] }
  E: { code_hash, container_digest, sbom_hash, cbom_hash,
      dataset_hashes[], device_ids[], seeds[], run_log_hash }
  V: [ { verifier_id, verifier_hash, method, input_hash, verdict,
      stats: {test, alpha, effect, ci}, independence_vector,
      operator, timestamp } ]
  status: { L1: bool, L2: bool, L3: bool, scope_verified,
          refuted: bool }
  P: { model: {provider, name, version|weights_hash},
      prompt_template_ver, prompt_hash, sampling, transcript_hash,
      data_handling: local|onprem|third_party, analysts[],
      custody_events[], legal_authority_ref,
      sth: {tree_size, root, signatures: [classical, ML-DSA|SLH-DSA]}

```

}
}

Appendix B. Novelty Audit

Search strategy. We searched arXiv, the IACR ePrint archive and IACR venues, ACM, IEEE, USENIX, NDSS, Springer, MDPI and Nature, IETF Datatracker, NIST CSRC and general web search, through 30 September 2026. Search terms covered: AI, machine-learning, neural, LLM, agentic, automated and autonomous cryptanalysis; PQC, ML-KEM, Kyber, ML-DSA, Dilithium, SLH-DSA, HQC and code-based cryptanalysis; ML side-channel; cryptographic digital forensics and post-quantum forensics; HNDL; crypto-agility and migration verification; proof-carrying security and AI verification; and cryptographic digital twin.

Table 16. Novelty audit.

Prior work	Year	Core contribution	Overlap with this paper	Difference	Remaining gap
CryptanalysisBench [4]	2026	191-task LLM cryptanalysis benchmark; game-based verification; recall vs. novelty tiers	Executable verification; contamination awareness	Ours: persistent claim objects, scope labels, independence, forensic link	Claims not preserved as evidence; no SIR
AI Grinding [5]	2026	Autonomous workflow with exact witnesses and controls	Witness + control = our L1 for EX	Ours: claim-form-specific verifiers, L2/L3, provenance log, risk link	Independent replication not structured
Anthropic HAWK/AES [1]	2026	AI-assisted design-level and reduced-round results; verification pipeline	Motivating case	—	Verification gap noted by commentators [6]
CrypFormBench [14]	2026	LLM formal-analysis benchmark	Formal methods + LLM	Ours checks the <i>claims</i> , not protocols	—
CryptoFormalEval [40]	2024	LLM + Tamarin protocol vulnerability detection	PR claims via Tamarin	Ours spans all claim forms	—
Kestrel [60]	2026	Static PQC detection in stripped binaries for forensics and migration	Layer 2 ID claims; migration assurance	Ours adds claims, grading, HNDL	—
Quantum-safe code auditing [59]	2026	LLM-assisted static analysis + quantum risk scoring	Layer 2; risk scoring	Ours: retrospective/prospective HNDL; graded claims	—
CARS [61]	2026	Crypto-agility readiness score	Migration assessment	Readiness vs. verification of migration behaviour	—
CycloneDX CBOM [62]	2026	CBOM standard	Twin serialization	CBOM lacks claim/assurance edges	Extension needed
PQ audit evidence [23]	2025	Q-integrity notions; re-signing, Merkle anchoring	Layer-6 log design	Ours applies it to cryptanalytic claims	—
QFRA [63]	2026	Quantum forensic readiness for hybrid networks	PQ forensic readiness	Different problem (quantum evidence objects)	—
Cyber digital twin [74]	2026	Quantum-safe network digital twin for threat forecasting	“Digital twin” term	Network-defence twin, no claims/HNDL/forensics	—
HNDL temporal model [13]	2025	Temporal HNDL risk with longevity and migration	Layer 4	Ours separates retrospective/prospective; calibration	—
Certificates of validity for AI [67]	2026	Cryptographic certificates that AI actions satisfy policy	“Proof-carrying” AI outputs	Policy compliance, not cryptanalytic truth	—

Prior work	Year	Core contribution	Overlap with this paper	Difference	Remaining gap
Agent tool-use binding [68]	2026	Reproducibility commitments, interaction ledger	Provenance of agent actions	Generic agents; no cryptanalytic semantics	—
Proof-carrying code [66]	1997	Code carries a checkable safety proof	Origin of “proof-carrying”	Applies to programs, not claims	—

What is new (to the extent of our search). Based on the literature reviewed through 30 September 2026, we did not identify:

- (i) a claim taxonomy that maps the *logical form* of AI-generated cryptanalytic claims to verifier types and independence requirements;
- (ii) a scope-inflation metric;
- (iii) an HNDL model that separates retrospective from prospective exposure for forensic triage;
- (iv) a framework that combines these with persistent, post-quantum-resilient forensic provenance and a cryptographic-estate twin.

What is integration. The Merkle log [19], [20], hybrid re-signing [23], CBOM [62], deterministic PQC detection [60], executable game verification [4] and the witness-plus-control discipline [5].

What needs experiments before novelty can be claimed as *useful*. Whether claim-form-specific verification actually reduces FCCR and SIR (H2, H6), whether evidence objects improve IRR (H3), and whether the two-component HNDL model changes triage decisions (H4).

Appendix C. Final Quality Check

Area	Item	Status
Literature	Current through 30 Sep 2026; primary sources prioritized; competing work included; novelty checked	Done. Several 2026 items are preprints or blogs and are flagged (§16)
Cryptography	Full vs. reduced distinguished; design vs. implementation distinguished; threat model; PQC parameters	Done (Tables 1, 3, 4, 5; §6; §11.7)
AI	Hallucination; contamination; generalization; adversarial AI; reproducibility	Done (Table 6; §12 Track T; §17)
Forensics	Chain of custody; integrity; provenance; independent verification; auditability	Done (§10, §18)
Methodology	RQs, hypotheses, datasets, baselines, ablations, metrics, statistics, limitations	Done (§11–§16)
References	No fabricated references; identifiers verified where available; claims mapped to sources	Done. See audit note below
Novelty	Prior work compared; contributions isolated; no “first-ever” claims	Done (Appendix B)

Citation audit note. Every reference was located in a primary or publisher listing during preparation (NIST CSRC, IACR ePrint/TCHES, ACM DL, Springer, MDPI, Nature, arXiv, IETF, author or project sites). Where we could not confirm a DOI, we give the ePrint or arXiv identifier or the URL instead of guessing a DOI. Discrepancies between the source drafts supplied for this study and the primary sources were resolved in favour of the primary sources. Three examples: the drafts stated the HAWK-256 reduction both as $2^{150} \rightarrow 2^{108}$ and as $2^{64} \rightarrow 2^{38}$, and we report Anthropic’s figures for HAWK-256; the Kyber NTT result is a 2025 CNN study (96.64%), not a 2026 Transformer study; and the ML-DSA public-template attack needed about 300 traces, not 7,400. Two items in the drafts could not be verified, a “PQC-IRFF” framework and an ML migration-prioritization framework, so they are excluded. Before submission the author should re-verify all preprints against their final published versions and add page numbers where the venue requires them.

Research Contribution Assessment

1. Core proposed contribution. An *evidence-carrying* framework for AI-assisted cryptanalysis with five parts:

- a claim taxonomy based on logical form and scope;
- separate assurance levels (L1 executable, L2 independently reproduced, L3 formally verified);
- the evidence object $C = (H, W, E, V, P)$ in a signed, post-quantum-resilient Merkle log;
- a cryptographic digital twin with a retrospective/prospective HNDL model that links graded claims to forensic triage and migration verification;
- a pre-registered evaluation protocol (EVC-Bench, eleven metrics including the Scope-Inflation Rate).

2. Evidence that it addresses a real gap. Commentators on the 2026 results describe verification as the bottleneck [6] and point to the independence problem [15]. A widely circulated claim of an ML-KEM break did not survive scrutiny [69], and recent ML-KEM incidents were implementation faults that could be mistaken for design flaws [31]. Existing systems verify claims inside research benchmarks [4], [5], protect evidence integrity [23] or inventory deployments [60], [62]. None of the work we reviewed joins these up.

3. Closest existing works. CryptanalysisBench [4]; AI Grinding [5]; Kestrel [60]; post-quantum audit evidence [23]; CrypFormBench [14]; temporal HNDL modelling [13].

4. Exact distinction. Unlike [4] and [5], the framework (a) chooses verifiers by claim form rather than by benchmark design, (b) requires ϕ -independent reproduction, (c) records and measures scope inflation, and (d) keeps claims as forensic evidence outside the research setting. Unlike [60] and [62], it records *claims against* deployed cryptography and their assurance level, not just what is deployed. Unlike [23], it applies post-quantum evidence integrity to *cryptanalytic inferences*. Unlike [13], it separates exposure that migration cannot reduce from exposure that it can.

5. What must still be demonstrated. Four effects must be demonstrated: that verification gates reduce FCCR and SIR in reports (H2, H6); that evidence objects improve IRR and time-to-reproduce (H3); that the HNDL model changes and improves triage against an expert or counterfactual ground truth (H4); and that the independence sets $E(\phi)$ capture the real correlated-failure channels.

6. Potential weaknesses.

- The contribution is mainly *integrative and methodological*. Reviewers at cryptography venues may see it as engineering rather than cryptanalysis.
- L3 may be impractical for most real CX claims.
- Ground truth for HNDL is modelled.
- Benchmarks with planted faults may overstate effectiveness.
- Fast change in frontier models limits how long the empirical results stay valid.
- The framework adds cost (VC), which practitioners may not accept unless it is quantified.

7. Most important experiments before publication. In order of priority:

1. **E5 + E7:** FCCR, SIR and IRR with and without the framework on CryptanalysisBench tasks plus sealed hallucination traps.
2. **E3:** planted-fault implementation targets with a controlled analyst study.
3. **E2:** HNDL triage against a blinded expert panel.
4. **E10:** a documented negative result on full-parameter standardized schemes using the Part-19 reporting template.

8. Recommended target venues. If E5/E7 and E2 are done:

- **Digital forensics:** DFRWS (USA or EU) or *Forensic Science International: Digital Investigation*.
- **Security systems and methodology:** ACM CCS, USENIX Security, or IEEE EuroS&P (systematization or methodology track, if the empirical study is strong).
- **Post-quantum:** a PQC workshop or a NIST PQC seminar for the migration-verification component.
- **SoK-style version:** if experiments are limited, the claim taxonomy, failure-mode table and verification decision table could be submitted as an SoK paper, for example to IEEE S&P's SoK track or IACR Communications in Cryptology.

9. Publication-readiness assessment. *Evidence-based, not a score.* The manuscript is a well-scoped framework and protocol paper without empirical validation. In its current form it is suitable as:

- a workshop paper;
- a pre-registered study proposal;
- the basis for an SoK submission, provided the related-work and failure-mode analysis are extended and peer-reviewed sources replace the 2026 preprints and blogs where possible.

It is **not ready** for a top-tier full-paper venue. Its main claims are effectiveness claims (H2–H4, H6), and no experiment supports them yet. The minimum for a full-paper submission is Phase 1 and Phase 2 of §14.3 with pre-registered analysis.